# When Does Ability Grouping Improve Learning? Evidence from Three Experiments*

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**Abstract**

Grade is a cheap but noisy signal of what students are ready to learn. Cross-grade ability grouping replaces that incumbent signal with a placement test, but the value of this substitution depends on more than sorting students correctly. We study three randomized evaluations of cross-grade grouping in Kenya, Liberia, and Nigeria in a scripted-instruction system where intended content is fixed by group. The observed implementations did not generate clearly positive average effects. The reason is economically informative: each setting was missing at least one complementary input. The placement test must identify the right level, schools must form and maintain the intended groups, and teachers must deliver the assigned lessons. A model of assignment and learning combines these experimental and implementation margins to recover the missing high-input case. The preferred benchmark predicts gains of about 0.19 standard deviations when all three inputs are jointly strong, a magnitude comparable to benchmark tracking gains under teacher discretion. Ability grouping is therefore not a yes-or-no policy. It is a signal-substitution technology whose payoff appears only when measurement, management, and instruction work together.

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# 1 Introduction

Organizations allocate heterogeneous people to production environments using imperfect signals. Firms assign workers to tasks, hospitals assign patients to units, lenders assign borrowers to contracts, and schools assign children to classrooms. In each case, the default rule is rarely neutral. It is usually a cheap incumbent signal: noisy, administratively convenient, and valuable only to the extent that it predicts the productive match.

In schools, the incumbent signal is grade. Grade-based assignment groups students by age and curricular exposure, then teaches each classroom at grade-level expectations. This rule has an obvious organizational logic: it is cheap, legible, and coordinated with curriculum sequencing. But grade is a proxy, not a primitive in the production of skill. Human capital theory treats learning as the product of skills and instructional inputs, not age labels (Ben-Porath, 1967); dynamic models of skill formation strengthen the point because early mismatch can compound when missing foundations lower the return to later instruction (Cunha and Heckman, 2007). Grade works well when progression through school is tightly coupled to learning. It works poorly when late enrollment, repetition, automatic promotion, and heterogeneous home inputs decouple grade from skill.

In many low-income school systems, grade and skill are only weakly coupled. Within-grade achievement variance is wide, adjacent grades overlap substantially in skill, and many students reach upper primary without mastering foundational literacy or numeracy (Banerjee et al., 2007; Muralidharan et al., 2019). A single nominal Grade 3 classroom may contain students who cannot read a sentence and others who can read fluently; in mathematics, some students cannot count while others are ready for multiplication. In such settings, the question is not whether to sort students. Grade already sorts. The question is whether another signal can improve on the grade rule.

Cross-grade ability grouping is the classic organizational response. The Joplin Plan, introduced in the 1950s, regrouped elementary students across grade lines for reading so that instruction could be pitched to measured reading level rather than age-grade placement (Floyd, 1954). Its logic is closely related to the instructional-level tradition in educational psychology: students learn most when content is neither trivially easy nor beyond reach (Betts, 1946; Vygotsky, 1978). The appeal is also economic. Cross-grade grouping expands the assignment margin relative to within-grade tracking, but it requires no additional teachers or classrooms. It changes who sits where, not the school budget.

The Joplin Plan evidence provides the historical benchmark. In a best-evidence synthesis, Slavin (1987) reports a median effect size of +0.45 SD for the Joplin Plan, and a related review of nongraded elementary schools finds similarly large gains for one-subject cross-grade grouping (Gutiérrez and Slavin, 1992). These estimates come from older studies and should not be read as a modern experimental benchmark. They are useful because they point to mechanism. Slavin (1987) finds grouping most effective when teachers can make use of the regrouped classroom: grouping is limited to one or two subjects, assignments are frequently reassessed, and teachers vary level and pace in response to student needs. The Joplin Plan's success was sorting plus adaptation, not sorting alone.

That distinction is the starting point for this paper. The standard tracking model, as in Duflo et al. (2011) and the classroom-production logic of Lazear (2001), works through teacher-mediated homogeneity. Tracking narrows the distribution of ability in the room, and the teacher adjusts content, pace, effort, or classroom management to that distribution. This is why peer composition, teacher response, and assignment are bundled in most tracking studies. Scripted instruction changes the production technology. If lesson content is fixed by track-specific scripts, grouping can no longer work mainly by inducing teachers to retarget instruction to a more homogeneous classroom. It works, if it works, by routing each student to the correct fixed content.

The change in technology transforms the sufficient statistic. Under teacher discretion, classroom homogeneity can be productive because it helps the teacher adapt. Under scripted instruction, homogeneity is not enough. A classroom can be homogeneous at the wrong level. What matters is whether the diagnostic places students into content they are ready to learn, whether the school actually forms those groups, and whether the assigned content is delivered. Scripted grouping therefore raises the return to correct placement while also raising the cost of misplacement. A teacher with discretion may partially accommodate a student placed too high; a scripted multiplication lesson cannot teach a student who has not learned to count.

We formalize this as a signal-substitution problem. Grade-based assignment is not the absence of tracking; it is tracking by age and curricular exposure. Cross-grade ability grouping substitutes a diagnostic signal for that incumbent signal. The substitution is valuable only if the diagnostic rule reduces instructional mismatch more than the grade rule already does. The assignment margin is therefore comparative, not absolute. A diagnostic with high predictive content may add little in a tightly managed school system where grade progression already tracks learning; a more modest diagnostic may have higher value in a chaotic system where grade carries little information. If the diagnostic is highly informative and the grade signal is weak, the new rule can create large gains. If the diagnostic is noisy, the school may discard a useful grade signal and faithfully execute a worse assignment rule. This observation reframes null effects in the tracking literature: a null need not imply that grouping has no latent value. It may mean that the new signal failed to dominate the incumbent rule, or that the organization failed to implement the productive response to the signal.

The framework has three primitives. The first is diagnostic quality, $\rho$: how much information the placement signal contains about the ability that predicts future learning. This is a policy input, not a nuisance measurement parameter, and is closely related to the psychometric idea that a test's value depends on its reliability and validity for its intended use (Lord and Novick, 1968; Cronbach and Meehl, 1955). The second is assignment execution, $\omega$: how faithfully intended groups are formed and maintained. The third is assignment-channel delivery fidelity, $\tau$: how faithfully the assigned track's curriculum is actually delivered. Recent work on external validity emphasizes that realized null effects need not imply null latent effects when implementation is incomplete (Angrist and Meager, 2023). Ability grouping makes the point especially stark because implementation is not a scalar dose. The inputs are complements. A better diagnostic has little value if students are not assigned according to it, and correctly assigned groups have little value if the curriculum that makes those groups productive is not delivered.

We study this problem using three randomized evaluations of cross-grade ability grouping in schools operated by NewGlobe Education, a large school network that uses scripted lesson plans delivered through digital tablets. The experiments span Kenya, Liberia, and Nigeria; literacy and numeracy; and two- and three-track designs. The common operator matters. All three experiments took place within the same broad instructional architecture, but they are not identical treatments. We therefore do not treat the pooled effect as the main estimand. Instead, the experiments occupy different cells of the implementation space highlighted by the model, letting us ask why the same organizational idea produces different effects.

The three experiments produce muted average effects for different reasons, and each identifies a different missing input. Liberia is the low-signal cell. Assignment was mechanically executed, but the baseline diagnostic explains less than 7% of endline variation ($\rho = 0.055$), so the intervention replaced grade with an almost uninformative signal. Grouping reduces achievement by 0.21 SD ($p = 0.12$). This is not a small implementation dose applied to a positive latent treatment. It is faithful execution of a weak assignment rule. Kenya is the high-signal, high-execution, moderate-delivery cell. The diagnostic is highly predictive ($\rho = 0.53$), assignment execution is clean ($\omega = 1$), and the intervention sharply compresses within-class dispersion. Yet the average effect is near zero (0.031 SD, $p = 0.56$), students with larger predicted assignment gains do not benefit more, and movers do not outperform stayers. Kenya is therefore the fact the structural model must respect. Accurate sorting and clean execution are not enough at the observed level of delivery, especially in a setting where the incumbent grade signal was already comparatively informative.

Nigeria was designed to occupy the high-input cell. Lagos was chosen because general delivery quality was high, and the experiment was intended to implement a three-track numeracy design. In practice, the Yellow/top group was thin, so our preferred reduced-form specification collapses Nigeria to the support-aware two-group contrast Red versus Blue/Yellow, with unrestricted three-group estimates reported in the appendix. The higher-level Blue/Yellow group has a positive T3 endterm effect of +0.236 SD (SE = 0.132) and a stacked endterm effect of +0.214 SD (SE = 0.095). But implementation evidence shows that the intended high-input cell was not realized: assignment records are missing for about one-third of treated students, many schools failed to maintain the intended group structure, teachers frequently delivered content for the wrong track, and the Direct Instruction numeracy curriculum that carries the assignment-rule contrast was delivered in only 26% of scheduled lessons. The estimated assignment-execution primitive is $\omega = 0.84$ and delivery fidelity is $\tau = 0.31$.

We use three kinds of evidence. Random assignment identifies the realized effects of the implemented policies. Implementation and process data explain why those effects differ across settings. A structural model combines these margins to study the benchmark that no experiment directly observes: the high-signal, high-execution, high-delivery case. Because no experiment jointly achieves high $\rho$, high $\omega$, and high $\tau$, the reduced form cannot answer the main counterfactual question: what would scripted cross-grade grouping produce if the productive inputs were jointly present? The preferred benchmark is concrete: Kenya-level signal quality, near-clean assignment execution, and high assignment-channel delivery fidelity. It predicts an ATE of about +0.19 SD. This magnitude is in the range of the 0.14–0.18 SD gains in Duflo et al. (2011) under teacher discretion and is conservative relative to the older Joplin Plan record. The observed experiments do not replicate those benchmarks. They show that gains of this size are plausible under scripted cross-grade grouping only when accurate assignment is paired with very high delivery fidelity.

The closest economics literature studies tracking and ability grouping. Meta-analyses of within-grade tracking generally find small or mixed average effects (Slavin, 1987, 1990; Kulik and Kulik, 1987; Terrin and Triventi, 2022), with important heterogeneity across tracks and institutional settings (Burke and Sass, 2013; Hoxby and Weingarth, 2005; Antecol et al., 2016; Garlick, 2018; de Roux and Riehl, 2022). In developing countries, the main modern benchmark is Duflo et al. (2011), who find gains from within-grade tracking in Kenya under teacher discretion. Recent evidence from other settings similarly suggests that grouping can raise achievement for some students while leaving open why effects differ across environments (Busso and Frisancho, 2023). We study a different assignment margin: cross-grade grouping, the form historically associated with the Joplin Plan (Floyd, 1954; Slavin, 1987; Gutiérrez and Slavin, 1992).[1] This margin is more than a stronger version of within-grade tracking. It replaces grade with a diagnostic signal, so the economic question is whether the new signal improves on the assignment rule already in use.

The signal-substitution view helps reconcile the equity critique with the efficiency evidence. Work in sociology and education emphasizes that tracking can stigmatize lower-track students, reduce teacher expectations, and concentrate weaker instruction in lower groups (Oakes, 1985; Gamoran, 1992; Gamoran and Nystrand, 1994). These are not side concerns; they are production channels. In our framework, negative effects arise when the placement signal is weak, when assignments are sticky or poorly managed, or when lower groups receive inferior content. The disagreement in the literature is therefore partly about implementation regimes, not simply about whether grouping is good or bad. A realized tracking policy can harm students in one institution and raise learning in another without implying that assigning students to appropriate instructional levels has the same latent value everywhere.

The argument also links ability grouping to educational production and targeted instruction.

[1] There is also a large body of research on between-school ability grouping, mainly from Europe where tracking between academic and vocational tracks is prevalent (Fu and Mehta, 2018; Lavrijsen and Nicaise, 2015). We focus on within-school ability grouping.

In Lazear (2001), students share teacher time, pace, and classroom order, so peer composition and instructional response enter the classroom production function. Tracking experiments usually move all of these margins at once: peers, ranks, teacher behavior, and content. Scripted instruction changes what grouping is meant to do. It makes the empirical object closer to an allocation problem: can the school use a signal to match students to the lesson they are ready to learn? This links the paper to work on peer effects and group assignment (Hoxby and Weingarth, 2005; Carrell et al., 2013; Vigdor and Nechyba, 2007; Rjosk, 2022), but it also changes the interpretation of the peer channel because the classroom input depends on the instructional technology.

Targeted-instruction interventions show the value of aligning content to assessed learning levels (Banerjee et al., 2017; Duflo et al., 2020; Muralidharan et al., 2019). Success for All combines structured reading instruction, frequent assessment, and cross-grade reading regrouping in a form closely related to the Joplin Plan (Cheung et al., 2021), and its implementation literature shows that teacher beliefs, adaptations, and local take-up shape whether the model is actually delivered (Datnow and Castellano, 2000). Our setting makes the point in a sharper allocation environment. A placement test is not targeted instruction. It becomes targeted instruction only if the test is valid for the decision, the organization acts on the signal, and the assigned content is delivered.

The paper also speaks to measurement and implementation. Human capital models treat learning as the production of skills from prior skills and inputs (Ben-Porath, 1967; Cunha and Heckman, 2007); in such a production process, a test is valuable only if it is valid for the assignment decision it is meant to inform. The psychometric distinction between measurement and use is therefore central: reliability and validity are not abstract properties of a score, but determine whether a score can support a particular decision (Lord and Novick, 1968; Cronbach and Meehl, 1955). Angrist and Meager (2023) emphasize that null realized effects need not imply null latent effects when implementation is incomplete or unobserved. We build on that insight by showing that implementation in allocation policies is multidimensional and nonseparable. Signal quality, assignment execution, and delivery fidelity are complementary inputs. A better diagnostic has little value if schools do not act on it, and faithful execution can be harmful when the signal is wrong.

Together, the evidence gives a constructive reading of three muted reduced-form results. The reduced form shows that the observed implementations did not generate clearly positive average effects. The model shows why that finding is not sufficient by itself: the experiments reveal different missing inputs in the same assignment technology, and the high-input benchmark predicts about +0.19 SD when measurement, management, and delivery are jointly strong.

Section 2 formalizes the signal-substitution logic and the complementarity among diagnostic quality, assignment execution, and delivery fidelity. Section 3 describes the three experiments and the institutional differences that make grade more or less informative across settings. Sections 4–5 report the reduced-form effects and the reallocation diagnostics. Section 7 estimates the structural model and the high-input counterfactual. Section 8 returns to the tracking debate and the policy question of when a diagnostic should replace grade.

## 2 Theoretical Framework

The model starts from the distinction between the effect of the implemented policy and the effect the same assignment technology could produce under stronger implementation. That distinction is necessary because the three experiments do not all test the same point in the production space. Liberia tests faithful execution of a weak signal; Kenya tests accurate sorting with only moderate delivery; Nigeria was designed as the high-input case but did not execute or deliver the assigned curriculum. Following the model-first logic in Duflo et al. (2011), the theory and reduced form are organized around the same objects: the overall ITT, the sorting first stage, effects by track position, residual peer/rank diagnostics, and the counterfactual effect when the productive inputs are jointly present.

The economic problem is an allocation problem. Students are enrolled in grade-based classrooms, but their actual preparedness may differ sharply from grade level. Ability grouping can raise achievement if it moves students closer to instruction they can learn from. It can also fail, or even harm students, if the diagnostic signal is poor, if the intended groups are not implemented, or if the assigned curriculum is not delivered. The framework therefore separates three primitives: diagnostic quality ($\rho$), assignment execution ($\omega$), and assignment-channel delivery fidelity ($\tau$). The first primitive measures whether the school knows where students should be placed; the second measures whether that information is translated into realized groups; the third measures whether the assigned track's content is actually taught.

## 2.1 Realized Effects and Implementation

Angrist and Meager (2023) show that a null realized effect need not imply zero latent value when implementation is incomplete. In this setting, the implementation problem is not a single dose. The return to any one input depends on the others. Faithfully executing a weak diagnostic can place students into the wrong content. Faithfully delivering track-specific lessons has high returns only when students have been assigned to the right track. Better diagnostics matter most when assignment execution and delivery fidelity are high.

Let $m$ index a market. We write the realized treatment effect as

$$\mathrm{RTE}_m = \mathcal{P}_m(\rho_m, \omega_m, \tau_m; X_m), \tag{1}$$

where $X_m$ collects the market's classroom environment and production shifters. The target is not a treatment effect purged of context, but the production map $\mathcal{P}_m(\cdot)$ evaluated at alternative implementation regimes. The high-input counterfactual is

$$\mathrm{ATE}_m^H(\rho^H, \omega^H, \tau^H) = \mathcal{P}_m(\rho^H, \omega^H, \tau^H; X_m), \tag{2}$$

the high-implementation effect of the same organizational technology when diagnostic quality, assignment execution, and delivery fidelity are jointly high.

This formulation differs from a scalar implementation adjustment. The primitive $\rho$ measures targeting: whether the placement signal contains information about ability. The primitives $\omega$ and $\tau$ measure two distinct implementation margins. The first is execution of the assignment rule; the second is delivery of the instructional response. Because these inputs are complements, the model can rationalize an observed null without concluding that assignment-to-level is unproductive.

## 2.2 Environment and Assignment

Student $i$ in grade $g(i) \in \{1, \ldots, G\}$ has latent ability

$$\theta_i = \mu_{g(i)} + \nu_i, \tag{3}$$

where $\mu_{g(i)}$ is the grade-specific mean and $\nu_i$ captures within-grade heterogeneity, with $\mathbb{E}[\nu_i] = 0$ and $\mathrm{Var}(\nu_i \mid g(i)) = \sigma^2_{\nu,g}$. The school observes a noisy diagnostic score

$$s_i = \theta_i + u_i, \tag{4}$$

with $\mathbb{E}[u_i \mid \theta_i, g(i)] = 0$ and $\mathrm{Var}(u_i \mid g(i)) = \sigma^2_{u,g}$. The posterior expectation of ability given the signal and grade is

$$a_i \equiv \mathbb{E}[\theta_i \mid s_i, g(i)] = \mu_{g(i)} + \rho_g(s_i - \mu_{g(i)}), \tag{5}$$

where

$$\rho_g \equiv \frac{\sigma^2_{\nu,g}}{\sigma^2_{\nu,g} + \sigma^2_{u,g}} \tag{6}$$

is the signal quality of the diagnostic. Our empirical proxy is $R^2_g$, the within-grade $R^2$ from regressing endline scores on baseline scores in the control group. This is a policy predictive-content measure rather than a pure psychometric reliability statistic: it captures whether the baseline diagnostic improves assignment for the future outcome the intervention is trying to move, relative to the grade-based rule already in place.

The policy estimand is the diagnostic's incremental value over the incumbent grade rule. Let $\rho_{0m}$ denote the predictive content of grade-based assignment in market $m$. Then the assignment margin depends on the diagnostic's incremental predictive content, $\Delta\rho_m = \rho_m - \rho_{0m}$, not on $\rho_m$ alone. When grade progression is tightly coupled to learning because the operator controls enrollment, promotion, and placement, $\rho_{0m}$ is high and the diagnostic must clear a higher bar to generate assignment gains. When grade progression is weakly coupled to learning because of late enrollment, repetition, automatic promotion, or weak system management, $\rho_{0m}$ is low and an informative diagnostic has more to add. We do not add $\rho_0$ as a separately estimated primitive in the structural model, but the distinction matters for interpretation: high diagnostic quality need not imply a large treatment effect if the grade signal it replaces is already doing useful work.

Under the status quo regime ($G$), students remain in grade-based classrooms and receive grade-level instruction $I_i^G = I_{g(i)}$. Under score-based grouping ($T$), the school assigns students to one of $K \geq 2$ instructional tracks:

$$k_i^T \in \{1, \ldots, K\}, \qquad I_i^T = I_{k_i^T}, \tag{7}$$

where $I_1 < I_2 < \cdots < I_K$ are the instructional levels associated with each track. Our experiments implement $K = 2$ (Kenya, Liberia) and $K = 3$ (Nigeria). An important special case arises when $K = G$: the number of tracks equals the number of grades, the school creates the same number of classrooms with approximately the same instructional levels, and only the assignment rule changes. The class-size margin is approximately neutralized. All three experiments implement $K = G$: Kenya and Liberia each have $K = G = 2$ (Liberia runs two parallel sub-experiments, one for Grades 1–2 and one for Grades 3–4, each with $K = G = 2$); Nigeria has $K = G = 3$.

The intended track assignment $k_i^*$ is determined by the diagnostic score and a set of cutoffs. We separate this from *realized* track assignment, which may differ due to implementation frictions. Let $\omega \in [0, 1]$ denote assignment execution: the degree to which the intended group structure is actually realized in the classroom. In Kenya and Liberia, assignment is deterministic ($\omega = 1$): the operator's protocols translated scores into groups mechanically. In Nigeria, the intended three-group structure collapsed in the majority of treatment schools. The structural model (Section 7) parameterizes this through a school-level implementation regime and misclassification matrix, with $\omega$ summarizing the overall execution quality. The separation of $\omega$ from $\rho$ is essential: $\rho$ measures whether the *signal* contains information about ability; $\omega$ measures whether an informative signal is *implemented* as the intended assignment. Nigeria has moderate $\rho$ (the diagnostic has real predictive power) but low $\omega$ (that information was not used).

## 2.3 Production Technology

The standard classroom-production logic begins with the observation that instruction is partly a shared input. In Lazear (2001), class size matters because each student can reduce the amount of productive classroom time available to others; more generally, the distribution of students shapes teacher attention, disruption, and pace. The standard tracking model—as in Duflo et al. (2011)—uses this production logic to explain why homogeneity can be valuable. The teacher targets instruction to the class composition, typically the median student. Under this technology, tracking works by creating homogeneous classes: when the within-class distribution narrows, the teacher's targeting is closer to each student, and all students benefit. Peer effects are inherent to the mechanism, because the channel through which homogeneity helps *is* the teacher's response to who is in the room.

Our setting uses a different source of variation. All three experiments take place in schools using scripted lesson plans delivered through digital tablets. The instructional model is designed to make intended content a function of track designation, not classroom composition. This feature is useful for interpretation: relative to teacher-discretion tracking, the design attenuates the endogenous retargeting response and focuses the analysis on whether students are assigned to the right content and whether that content is delivered. We parameterize the strength of this restriction with $\tau \in [0, 1]$: assignment-channel delivery fidelity.

Endline achievement under assignment regime $a \in \{G, T\}$ is

$$Y_i(a) = \theta_i + \alpha - \lambda(\tau)\big(\theta_i - I_i(a)\big)^2 + \pi(\tau)\,\bar{\theta}_{-i,c(i,a)} + \varepsilon_i(a), \tag{8}$$

where $\bar{\theta}_{-i,c(i,a)}$ is the mean ability of student $i$'s classmates under regime $a$. The term $(\theta_i - I_i(a))^2$ is instructional mismatch: a student whose ability is far from the instructional target learns less. The penalty $\lambda(\tau)$ is increasing in assignment-channel delivery fidelity,

$$\frac{\partial \lambda}{\partial \tau} > 0. \tag{9}$$

When the teacher follows the track-specific script, instruction is determined by track designation and cannot easily adjust to the student. A student who cannot count receives little benefit from a multiplication lesson, even if that lesson is well delivered. When the teacher exercises discretion, she may partially accommodate the student by slowing the pace or revisiting prerequisites. Scripted instruction therefore raises the return to being in the correct track, but it also raises the cost of being in the wrong track.

Scripted assignment changes the sufficient statistic for the return to grouping. Under teacher discretion, the DDK mechanism makes within-class homogeneity productive because teachers adapt instruction to the class. Under scripted assignment-to-level, the key statistic is correct assignment: grouping helps when students hear the content that fits their preparedness. These objects can diverge. A class can be heterogeneous but correctly assigned to the right level of content, or homogeneous but assigned to the wrong level.

The two mechanisms can be nested. In a teacher-discretion environment, grouping can add a homogeneity-adaptation term, say $\eta A_m H_c$, where $H_c$ measures the homogeneity of classroom $c$ and $A_m$ measures the scope for teachers to adapt instruction to that homogeneity. In a DDK-style setting, $A_m$ is high: tracking changes classroom composition and teachers have discretion to respond. In the scripted setting studied here, $A_m$ is constrained by design, so the analogous productive margin is assignment to the correct content and faithful delivery of that content. This is why the DDK evidence is a high-adaptation benchmark for the same broad intervention family, not a literal fourth cell in the cross-grade assignment technology.

Peer composition can still matter through the instructional environment: teachers adjust pace, stronger students model skills, and disruption or motivation may change classroom time use. But higher $\tau$ reduces the scope for peers to affect what gets taught through instructional targeting,

$$\frac{\partial \pi}{\partial \tau} \leq 0. \tag{10}$$

In the limit of perfectly scripted instruction, peer ability has no effect on own learning through teachers' choice of instructional level. This does not require classmates to have no effect. Rank, disruption, motivation, and informal help can still operate outside the targeted-content channel, and in practice $\tau < 1$. The attenuation is a matter of degree, which is why we treat peer/rank estimates as residual classroom-production diagnostics rather than as the paper's primary causal object.

Equation (8) subsumes both peer-level effects and rank effects into the $\pi(\tau)\bar{\theta}$ term. Regrouping changes the level of peers and the student's relative standing within the classroom, and these move in opposite directions. We cannot separately identify them. The Borusyak-Hull estimate $\hat{\zeta}$ (Section 4.3) is therefore interpreted as a composite peer/rank margin, not as the structural peer effect that drives the paper.

### 2.4 Implications for the Reduced Form

Define student $i$'s *assignment gain* as

$$B_i \equiv \left(\theta_i - I_{g(i)}\right)^2 - \left(\theta_i - I_{k_i}\right)^2. \tag{11}$$

The realized treatment effect for student $i$ decomposes into three channels:

$$\Delta Y_i = \underbrace{\lambda(\tau)\, B_i}_{\text{assignment gain}} + \underbrace{\pi(\tau)\, \Delta\bar{\theta}_i}_{\text{peer/rank}} + \underbrace{\text{organizational costs}}_{\text{grade mixing, implementation}} + \tilde{\varepsilon}_i, \tag{12}$$

where $\Delta\bar{\theta}_i \equiv \bar{\theta}_{-i,c(i,T)} - \bar{\theta}_{-i,c(i,G)}$. The second and third terms are the Lazear-style classroom-production margins: classmates affect learning through the instructional environment, and reorganization can change congestion, disruption, grade mixing, and the routines through which lessons are delivered. Averaging over students:

$$\text{ATE}(\rho, \omega, \tau) = \lambda(\tau)\, \mathbb{E}[B_i(\rho, \omega)] + \pi(\tau)\, \mathbb{E}[\Delta\bar{\theta}_i] + \mathbb{E}[\text{org. cost}_i], \tag{13}$$

where $\omega$ governs how much of the intended assignment gain is implemented: $\mathbb{E}[B_i(\rho, \omega)]$ is increasing in both $\rho$ (better signal) and $\omega$ (more faithful execution).

The decomposition gives three empirical implications. First, the ITT is informative but not sufficient. A small ITT can arise because the assignment technology has low value, because the signal is weak, because the intended groups are not executed, or because the assigned content is not delivered. Second, treatment effects by track position are informative about whether the assignment rule is moving students toward productive content. With $K = G$, the largest potential assignment gains should be in the extreme tracks, because those tracks draw students whose preparedness is most different from their enrolled grade. Third, peer/rank diagnostics are useful mainly as a check on residual classroom-production channels. Under scripted instruction, the paper should not stand or fall on a causal peer-effect interpretation.

The key comparative static is complementarity between $\rho$, $\omega$, and $\tau$. The expected assignment gain $\mathbb{E}[B_i(\rho, \omega)]$ is increasing in signal quality and assignment execution: better sorting places more students in the correct track, and faithful execution makes the intended assignment real. The mismatch penalty $\lambda(\tau)$ is increasing in delivery fidelity: scripted instruction amplifies both the reward for correct placement and the cost of misplacement. Therefore:

$$\frac{\partial^2 \text{ATE}}{\partial\rho\, \partial\tau} = \underbrace{\frac{\partial\lambda}{\partial\tau}\, \frac{\partial\mathbb{E}[B_i]}{\partial\rho}}_{>0} + \underbrace{\frac{\partial\pi}{\partial\tau}\, \frac{\partial\mathbb{E}[\Delta\bar{\theta}_i]}{\partial\rho}}_{\text{ambiguous, typically small}} > 0. \tag{14}$$

The first term is positive: the return to better sorting is amplified when content is faithfully delivered. The second term depends on how sorting changes peer composition and on $\partial\pi/\partial\tau \leq 0$; it is typically small because better sorting need not have a systematic directional effect on the average peer shift. Assignment execution $\omega$ is a prerequisite for this complementarity. If intended assignments are not realized, better diagnostics and better scripted content cannot interact productively.

**Remark 1** (Who benefits from $K = G$ grouping)**.** *When $K = G$ and $I_k \approx I_g$, the expected assignment gain $\mathbb{E}[B_i \mid k_i = k]$ is largest for extreme tracks ($k = 1$ and $k = K$) and smallest for middle tracks. This follows because extreme tracks draw disproportionately from students whose ability is most discrepant from their enrolled grade level. With $K = G = 3$, effects should be largest for the top and bottom tracks and smallest for the middle track.*

### 2.5 Mapping the Three Experiments

The operator's pre-experiment administrative data provide an ordinal ranking of *general* delivery quality across settings (Table 1): Liberia < Kenya < Nigeria. This is the ranking that motivated the Nigeria experiment. But the framework's $\tau$ refers specifically to assignment-channel delivery fidelity—how faithfully the curriculum through which the assignment-rule difference operates is actually delivered. As we document in Section 3, this assignment-channel $\tau$ does not follow the ex ante ranking: the DI numeracy curriculum in Nigeria was delivered in only 26% of scheduled lessons, while the literacy revision in Kenya and Liberia was delivered through the established curriculum. The structural estimates (Section 7) recover $\hat{\tau}_{\text{Kenya}} = 0.50 > \hat{\tau}_{\text{Liberia}} = 0.38 > \hat{\tau}_{\text{Nigeria}} = 0.31$.

Signal quality $\rho$ is measured within each experiment by baseline predictive power: $\hat{\rho}_{\text{Kenya}} = 0.53$, $\hat{\rho}_{\text{Nigeria}} = 0.12$, $\hat{\rho}_{\text{Liberia}} = 0.055$. Assignment execution $\omega$ is deterministic in Kenya and Liberia ($\hat{\omega} = 1$) and materially lower in Nigeria ($\hat{\omega} = 0.84$). The institutional discussion above adds an important interpretation: Kenya has the strongest diagnostic, but also the most informative incumbent grade signal. The incremental signal-substitution margin is therefore smaller than the raw value of $\rho$ alone would suggest.

The qualitative predictions follow directly. In Liberia, the lowest-$\rho$ setting, sorting should generate minimal assignment gains regardless of execution; with low assignment-channel delivery as well, the ATE should be near zero or negative. In Kenya, the high-$\rho$, high-$\omega$ setting, the model predicts a strong reallocation first stage and potentially valuable assignment margins, but the realized ATE depends on whether the diagnostic adds enough over an already informative grade signal and whether moderate delivery fidelity is enough to convert correct placement into learning once regrouping also changes peers, rank, class size, and grade composition. Nigeria was designed to test the high-$\rho$, high-$\omega$, high-$\tau$ case, but the experiment did not deliver the intended conditions. It therefore becomes evidence on implementation complementarity rather than a clean observation of the high-input cell. The structural model in Section 7 uses this cross-experiment variation to recover the missing counterfactual.

The evidence follows the same structure. When a statistic can be harmonized across all three experiments, we report all three together. When the design or data differ, we either report country-specific tables side by side or state the restriction explicitly. Kenya and Liberia share the 2018 two-track literacy design and therefore appear together in several diagnostic and robustness tables. Nigeria has a separate three-track numeracy design, two follow-up waves, and implementation audits, so the analogous Nigeria diagnostics are reported in adjacent Nigeria-specific tables. The structural counterfactual tables use all three experiments for observed-cell fit and primitive estimation, but the policy counterfactual rows focus on the two missing cells the design most directly informs: Kenya with higher assignment-channel delivery and Nigeria with improved execution or delivery. Liberia's role is the observed low-signal, high-execution cell.

# 3 Experimental Design and Data

## 3.1 Setting

All three experiments take place in schools operated by NewGlobe Education. NewGlobe operates large networks of low-cost schools using a standardized model: scripted lesson plans delivered through digital tablets, with teacher compliance monitored by school managers. The common production environment makes the experiments analytically comparable: each intervention changes the assignment rule within a system where intended lesson content is centrally specified.

The experiments share this organizational model but take place in settings that differ in how effectively it is executed. The operator's own administrative monitoring data, collected routinely prior to the experiments, document these differences. Table 1 reports the operator's pre-experiment assessment of lesson completion and teacher quality across settings. We treat these data as the ex ante delivery ranking that motivated the Nigeria design, and then distinguish it from assignment-channel delivery fidelity below.

They also differ in the quality of the incumbent grade signal. The Kenya experiment took place in Bridge/NewGlobe private academies, where the operator directly managed schools and had greater span of control over enrollment, grade placement, and progression. In that environment, grade is likely to be a comparatively informative organizational signal: students are more likely to be in the grade that the system believes matches their learning history. Liberia was different. NewGlobe managed public schools under a charter-style partnership in a system with much weaker baseline performance, so grade membership was less tightly connected to actual skill. Nigeria was also implemented in public schools, in Lagos, a stronger setting than Liberia but still not the same organizational environment as the Kenya Bridge academies. This ordering is useful for interpretation. A diagnostic can add less in Kenya precisely because the incumbent grade rule was less broken; in Liberia and Nigeria the potential value of replacing grade was larger, but only if the diagnostic and implementation system were strong enough.

Table 1: Operator's Pre-Experiment Assessment of General Delivery Quality

| | Liberia | Kenya | Nigeria (Lagos) |
|---|---|---|---|
| Lesson completion | Low | Medium | High |
| Teacher quality | Low | Medium | High |
| School audit pass rate | — | 19% | — |
| Programme quality | Status quo | Status quo | DI revamp |

*Notes:* Rankings are from the operator's internal experiment proposal (Sullivan, 2020), written before the Nigeria experiment. "DI revamp" indicates that both treatment and control arms in Nigeria received a newly developed Direct Instruction numeracy curriculum; prior experiments used the operator's existing ("status quo") literacy programme. The Kenya school audit pass rate (19% "perfect or passing") is from the same document. The Nigeria experiment was selected because Lagos had the highest lesson completion and teacher quality among the operator's markets. This table reports the *ex ante* general delivery quality ranking. Assignment-channel delivery fidelity ($\tau$)—the fidelity with which the curriculum carrying the assignment-rule difference is delivered—does not follow this ranking: the DI numeracy programme in Nigeria achieved only 26% lesson completion (Table 2), well below the established curriculum's 69%. The structural estimates (Section 7) recover $\hat{\tau}_{\text{Kenya}} = 0.50 > \hat{\tau}_{\text{Liberia}} = 0.38 > \hat{\tau}_{\text{Nigeria}} = 0.31$.

## 3.2 Experiment 1: Kenya (2018)

In August 2018, 50 out of approximately 300 Bridge/NewGlobe academies in Kenya were randomized to receive the reading group intervention. In treated schools, students in Grades 1 and 2 ($G = 2$) were assigned to one of two reading groups ($K = 2$; "Blue" and "Purple") based on a combined literacy and English baseline test administered in June 2018. Grade 1 students scoring above 40 points were assigned to the Purple (upper) group; Grade 2 students above 35 points.[2] Treatment was stratified by constituency. Students received 90 minutes of instruction with their reading group daily for Term 3 (approximately 3 months). Because $K = G = 2$, the class-size margin is approximately neutralized.

## 3.3 Experiment 2: Liberia (2018)

Approximately 90 public schools in Liberia, managed by NewGlobe under the Liberia Education Advancement Program, were randomized into treatment and control. The experiment consists of two parallel sub-experiments. In the first, students in Grades 1 and 2 ($G = 2$) were pooled and sorted into two tracks ($K = 2$) based on a baseline reading assessment; the cutoff for the upper track was 23 points. In the second, students in Grades 3 and 4 ($G = 2$) were similarly pooled into two tracks with a cutoff of 14 points. Each sub-experiment has $K = G = 2$: two grades of students sorted into two ability-based tracks. Treatment was stratified by grade group and treatment probability.

## 3.4 Experiment 3: Nigeria (2021)

The Nigeria experiment was explicitly designed to fill the high-$\rho$, high-$\omega$, high-$\tau$ cell. The operator's internal review noted that "operational challenges in the formal evaluations prevent us from understanding the true impact of a programme when implemented effectively and with fidelity."[3] Nigeria was selected because the operator's *general* delivery quality was highest there (Table 1); the placement protocol was designed to make $\rho$ and $\omega$ as high as possible, with School Supervisors administering one-on-one diagnostic exams and creating classroom rosters.

[2] Cutoffs were chosen by NewGlobe before the experiment to approximate an equal split of each grade's baseline score distribution. The grade-specific thresholds reflect differences in the raw-score distributions across grades.

[3] From the operator's internal experiment proposal (Sullivan, 2020). The proposal also noted that Lagos had the highest lesson completion and teacher quality among the operator's markets.

In January 2021, 42 NewGlobe academies in Lagos were randomized into treatment ($n = 20$) and control ($n = 22$), stratified by constituency. Both arms received the same three-level Direct Instruction (DI) numeracy curriculum: Level A (approximately P1 content), Level B (approximately P2), Level C (approximately P3). In control schools, the DI curriculum was assigned by grade. In treatment schools, students in Grades 1–3 ($G = 3$) were sorted into three ability-based tracks ($K = 3$)—Red (bottom, Level A), Blue (middle, Level B), and Yellow (top, Level C)—based on an 8-question two-stage placement exam. For outcome heterogeneity, however, the realized data support an effective two-track specification better than a separately estimated Yellow production cell. Yellow is sparsely populated in the realized rosters, so our preferred grouped reduced-form analysis compares Red to Blue/Yellow; one unrestricted Red/Blue/Yellow table is reported in the appendix for transparency.

The treatment contrast is purely the *assignment rule*: ability-based versus grade-based assignment to the same set of leveled curricula. Because $K = G = 3$, the three treatment classrooms replace the three grade-level classrooms, approximately preserving total class sizes (30.2 in control, 27.2 in treatment).

The operational data show that the Nigeria treatment did not disrupt routine school inputs, but that the curriculum carrying the assignment-rule contrast was delivered at low fidelity. Table 2 reports operational outcomes by treatment status.

Table 2: Nigeria Operational Outcomes by Treatment Status

| | Treatment | Control | $p$-value |
|---|---|---|---|
| Teacher attendance (%) | 93 | 92 | 0.68 |
| Pupil attendance (%) | 64.5 | 64.0 | 0.88 |
| Academy manager attendance (%) | 75 | 77 | 0.71 |
| Pupil-teacher ratio (T2) | 20.4 | 21.8 | 0.54 |
| Pupil-teacher ratio (T3) | 20.7 | 23.1 | 0.33 |
| DI Numeracy lesson completion (%) | 26.2 | 26.1 | 0.98 |
| Math lesson completion (%) | 69.4 | 68.7 | 0.85 |

*Notes:* Treatment-control differences in operational metrics from NewGlobe administrative data. DI Numeracy is the newly introduced Direct Instruction numeracy programme; Math is the pre-existing mathematics curriculum. Lesson completion is the fraction of scheduled lessons recorded as delivered. $p$-values are from regressions with strata fixed effects and academy-level clustering.

Operational metrics are balanced across arms, so the treatment did not disrupt the school's normal operations. The problem is instead the new curriculum. The DI numeracy curriculum was delivered in only 26% of scheduled lessons in both arms, while the pre-existing math curriculum maintained 69% completion. The new curriculum had not been absorbed into regular practice. Nigeria therefore ranks highest on the operator's *general* delivery-quality metrics (lesson completion and teacher quality for the existing curriculum), but the DI numeracy programme through which the assignment-rule difference is supposed to operate had low realized fidelity. Both arms were equally affected, so this is not a treatment effect on delivery but a system-level constraint on the DI programme. The 26% of lessons that were delivered provide some scope for the treatment to operate, and we exploit the remaining variation in the results.

By contrast, in Kenya, lesson-completion point estimates are positive but imprecise. The largest school-clustered estimate is 2.4 percentage points (SE = 0.017; Table A26), so the data do not support a large delivery disruption but also do not establish a precise positive delivery effect.

The Nigeria experiment was also *designed* to have high sorting accuracy, but phone-call audits and roster data reveal that it did not. Audits conducted throughout the intervention (61 calls across all 20 treatment academies) document substantial departures from the three-group design, including schools operating fewer than three groups and teachers delivering content for the wrong track. The placement test was not binding in realized rosters: in the cleaned analysis data, the Spearman rank correlation between placement exam scores and realized group assignment is only 0.25, and 34% of treatment students have no group assignment recorded. External audit summaries are even more adverse on some dimensions, reporting a Spearman correlation of 0.17 and fewer schools with a Yellow top track. Because these implementation sources conflict in levels, the structural model uses computed roster moments for estimation and keeps the external audit targets as validation moments.

The Nigeria design aimed to achieve high $\rho$, high $\omega$, and high $\tau$ simultaneously. It failed on two of three dimensions: assignment execution collapsed ($\omega = 0.84$), and assignment-channel delivery fidelity was low ($\tau = 0.31$, reflecting the 26% DI numeracy completion). The signal itself had moderate quality ($\rho = 0.12$), but even this information was not translated into the intended group structure. The structural model (Section 7) separates these failures and asks which margin binds.

### 3.5 Design Summary

Table 3: Three-Experiment Design Summary

| | Liberia | Kenya | Nigeria |
|---|---|---|---|
| Year | 2018 | 2018 | 2021 |
| Subject | Literacy | Literacy | Numeracy |
| Grades per sub-experiment ($G$) | 2 | 2 | 3 |
| Tracks ($K$) | 2 | 2 | 3 |
| $K$ vs. $G$ | $K = G$ | $K = G$ | $K = G$ |
| Schools (T / C) | ∼45 / ∼45 | 50 / 250 | 20 / 22 |
| *Three primitives* | | | |
| Signal quality ($\hat{\rho}$) | 0.055 | 0.533 | 0.121 |
| Assignment execution ($\hat{\omega}$) | 1.00 | 1.00 | 0.84 |
| Assignment-channel fidelity ($\hat{\tau}$) | 0.38 | 0.50 | 0.31 |
| Ex ante delivery ranking | Low | Medium | High |
| *Sorting accuracy detail* | | | |
| Incremental $R^2$ | 0.03–0.07 | 0.51–0.56 | 0.11–0.22 |
| Placement binding | Deterministic | Deterministic | Spearman = 0.17 |
| Delivery process moment | +1.1pp | +2.4pp | 26% DI completion |

*Notes:* Liberia runs two parallel sub-experiments (Grades 1–2 and Grades 3–4), each with $K = G = 2$. $\hat{\rho}$ is estimated from control-group baseline–endline $R^2$; $\hat{\omega}$ and $\hat{\tau}$ are from the structural model (Section 7), estimated from audit/roster and curriculum-specific delivery moments, respectively. "Ex ante delivery ranking" is the operator's pre-experiment assessment based on general lesson completion and teacher quality. The delivery process moment is the treatment-control difference in lesson completion for Kenya and Liberia, and the balanced DI numeracy completion level for Nigeria, where both arms received the new curriculum. Incremental $R^2$ is the within-grade variance explained by adding baseline scores to a grade-FE-only model.

### 3.6 Data

Administrative data were provided by NewGlobe for all three experiments. In Kenya, the baseline instrument is a combined literacy and English midterm examination. Administrative records also cover lesson completion rates and teacher attendance. In Liberia, the baseline is a reading pre-test. In Nigeria, the baseline is the Term 1 end-of-term maths score, and endline outcomes are percent-correct scores in maths and numeracy, standardized to mean zero and unit standard deviation in the control group. Nigeria's administrative data also include lesson-level completion records (40,692 observations), teacher attendance, pupil attendance, pupil-teacher ratios, and academy manager attendance.

Endline assessments were administered after the intervention in all three experiments. We standardize all scores to mean zero and unit standard deviation in the control group within each grade.

### 3.7 Balance and Attrition

Covariate balance is confirmed in all three experiments (Tables A4 and A5). In Kenya and Liberia, there is no differential attrition. In Nigeria, attrition is balanced at T2 but shows some imbalance at T3 endterm (treatment retains 65% vs. control 51%, $p = 0.12$); Lee bounds for the T3 endterm span $[-0.13, 0.30]$ (Table A38). The T2 endterm, with balanced attrition and tight Lee bounds of $[0.15, 0.18]$, serves as a cleaner benchmark.

## 4 Empirical Strategy

The empirical strategy follows the model's sequence. We first estimate the experimental ITT in each country, the quantity delivered directly by random assignment. We then ask whether the assignment rule created the reallocation the model requires: did the diagnostic predict later achievement, and did treatment change within-class dispersion and track position? Finally, we estimate residual peer/rank diagnostics. These diagnostics help interpret classroom production, but they are not the main causal object; the paper's main interpretation rests on the ITT, the assignment first stage, and the implementation primitives.

### 4.1 Intent-to-Treat Effects

For each experiment, the main estimating equation is

$$Y_i = \alpha + \beta T_j + \gamma \hat{\theta}_i^{EB} + \phi_s + \varepsilon_i \tag{15}$$

where $Y_i$ is the standardized endline score, $T_j$ is school-level treatment assignment, $\phi_s$ are strata fixed effects, and standard errors are clustered at the randomization level. The baseline control $\hat{\theta}_i^{EB}$ is an empirical Bayes ability measure that shrinks each student's raw score toward the grade mean by the grade-specific baseline predictive power:

$$\hat{\theta}_i^{EB} = \bar{s}_{g(i)} + \widehat{R}^2_{g(i)}\,(s_i - \bar{s}_{g(i)}), \tag{16}$$

where $\widehat{R}_g^2$ is the within-grade baseline–endline $R^2$ in the control group. Because $R_g^2$ is attenuated by endline noise, this implemented proxy shrinks more aggressively than the latent posterior in the model; it is therefore a conservative baseline ability control.

We report country-specific estimates as the primary reduced-form evidence. Pooled specifications are used for a narrower question: whether the three experiments reveal a common positive average effect once the data are stacked. The pooled individual-participant-data regressions allow country-specific baseline slopes and country-nested strata fixed effects, and the study-level meta-analytic estimates give each experiment its own first-stage estimate before averaging. We do not interpret the pooled coefficient as the structural target, because the model predicts heterogeneity across implementation regimes.

### 4.2 Effect on Within-Class Dispersion

The first-stage reallocation test asks whether treatment changes the composition of realized classrooms. We use baseline scores, which are predetermined, and estimate

$$|s_i - \bar{s}_{k(i)}| = \alpha + \beta T_j + \gamma \hat{\theta}_i^{EB} + \phi_s + \varepsilon_i. \tag{17}$$

A negative $\beta$ indicates that treatment compresses within-class baseline heterogeneity. Under the DDK homogeneity mechanism, this is the sufficient statistic for tracking's benefit. Under the assignment mechanism in our framework, it is informative but not sufficient. What matters is whether students are assigned to productive content, not only whether classmates are similar. This distinction helps interpret Nigeria, where realized treatment increases within-class dispersion even though the grouped outcome estimates are positive.

### 4.3 Peer/Rank Diagnostics

When a school is randomized into tracking, several classroom inputs change at once. A naive regression of outcomes on peer ability conflates peer composition with instructional match, class size, grade mixing, and teacher targeting. In a teacher-discretion environment, this simultaneity is part of the treatment. In our setting, scripted instruction gives a useful restriction for decomposition: intended content is assigned by track, so conditional on own score, treatment status, and track, instructional targeting is less directly chosen in response to realized peers.

The experiment then provides variation in classroom composition conditional on assigned content. Consider two students in different treated schools who have the same baseline score and are therefore assigned to the same track. Their intended instructional targets are identical under scripted curricula, but their classmates differ because the composition of each school differs. The challenge is that exposure is not randomly assigned within the realized track: a student's leave-one-out peer mean,

$$P_i = \frac{1}{n_{c(i)} - 1} \sum_{\ell \neq i,\, \ell \in c(i)} s_\ell, \tag{18}$$

depends on her own score through track assignment. Following Borusyak and Hull (2023), we remove this bias by conditioning on expected peer composition.

The diagnostic rests on three restrictions. Random assignment ensures that treatment is orthogonal to potential outcomes. Conditional independence requires that, conditional on own score and treatment status, classmates' baseline scores are not otherwise selected on the student's latent achievement shock. The scripted-instruction restriction requires intended instructional level to be a deterministic function of track designation, with deviations captured by assignment-channel fidelity $\tau$. This last restriction turns scripted instruction into an identification advantage rather than merely a narrow setting. When intended content is scripted, the mismatch measure $\hat{M}_i = (s_i - I_{D_i})^2$ is approximately a deterministic function of $(s_i, T_j)$ and is absorbed by score-bin $\times$ treatment fixed effects. At $\tau = 1$ this restriction holds exactly; at $\tau < 1$ it is an approximation, and residual teacher discretion or classroom dynamics can still matter.

**Assumption 1** (Random Assignment)**.** *$T_j \perp (\theta_i, u_i, \varepsilon_i)$ for all $i \in j$.*

**Assumption 2** (Conditional Independence)**.** *$\{s_k\}_{k \in j, k \neq i} \perp (\theta_i, \varepsilon_i) \mid s_i, T_j$.*

**Assumption 3** (Scripted Instruction)**.** *The intended instructional level $I_{kg}$ in each track $k$ is a deterministic function of track designation. Deviations from intended content are captured by assignment-channel delivery fidelity $\tau$.*

We implement the diagnostic using a control-function approach:

$$Y_i = \zeta P_i + \sum_{d,t,g} \delta_{dtg} \cdot \mathbb{1}\{D_i = d, T_j = t, G_i = g\} + \phi_s + \varepsilon_i. \tag{19}$$

The decile×treatment×grade cells absorb the expected peer composition, so $\zeta$ is identified from deviations around the cell-level expectation. The estimate $\hat{\zeta}$ captures the composite of peer-level and rank effects.

The model predicts that this social channel should be weaker as assignment-channel delivery fidelity rises, but the reduced-form $\hat{\zeta}$ also depends on how much structured peer-composition variation the treatment actually creates. We therefore use the peer estimates as diagnostic moments for residual classroom production rather than as a standalone test of $\tau$ or as the paper's main causal object.

# 5 Results

## 5.1 Experimental Effects

The results proceed in the order implied by the model. We begin with the experimental effects, then ask whether the assignment rule generated the intended reallocation, and finally examine residual classroom-production margins. The core reduced-form fact is simple: the observed experiments do not deliver a common positive effect, but the country pattern identifies which input is missing.

Table 15 presents the country-by-country ITT estimates using the primary endpoint for each experiment. Figure 1 summarizes the same experiments in the implementation space used by the model. In Liberia, grouping reduces endline scores by 0.212 SD ($p = 0.12$). In Kenya, the effect is near zero (0.031 SD, $p = 0.56$), with an upper confidence bound below the 0.18 SD found by Duflo et al. (2011). In Nigeria, the primary T3 endterm effect is positive but imprecise (+0.093 SD, SE = 0.133). Appendix Table A13 reports the Nigeria all-wave stacked estimate, +0.075 SD (SE = 0.076).

Because the theory predicts heterogeneity across implementation regimes, the pooled coefficient is not the paper's structural target. It answers a more limited question: is there a hidden positive average effect across the observed implementations? Table 16 shows that there is not. The student-weighted pooled estimate is $-0.070$ SD (SE = 0.071), the experiment-balanced estimate is $-0.058$ SD (SE = 0.077), and the fixed-effect meta-analytic estimate is +0.010 SD (SE = 0.046). The study-level heterogeneity diagnostics are moderate rather than decisive ($Q = 3.27$, $p = 0.195$, $I^2 = 39\%$). A parsimonious interaction with the observed primitive score $\rho\omega\tau$ is positive (0.683, SE = 0.494), but the highest observed score is still Kenya, where delivery fidelity is only moderate. The pooled analysis therefore rules out a simple common-effect success story; it does not rule out complementarity among inputs that are never jointly high in the observed data.

Table 17 reports treatment effects by track position. In Liberia, the negative effect is concentrated in the upper track ($-0.348$ SD, $p = 0.03$). Liberia executed the assignment rule, but the diagnostic was weak, so faithful execution assigned students to content that was not reliably appropriate for them. In Kenya, neither track shows significant effects. Accurate sorting and clean execution are therefore not enough, at observed delivery fidelity, to produce learning gains.

In Nigeria, the nominal three-color design creates a support problem for outcome heterogeneity. The Yellow group is the economically interesting top cell, but it is sparsely populated in realized rosters, with only 143 treated students in the T3 endterm sample. Our preferred grouped specification therefore collapses Blue and Yellow into a higher-level instructional group before estimation (Table 18). This is fixed by support and implementation logic, not by outcome patterns: the collapse preserves the main economic contrast between lower-level and higher-level instruction while avoiding overinterpretation of a thin nominal cell. In the preferred two-track specification, the Blue/Yellow effect is +0.236 SD at T3 endterm (SE = 0.132) and +0.214 SD in stacked endterm outcomes (SE = 0.095). The Blue/Yellow-minus-Red contrast is positive at T3 endterm (+0.194 SD, SE = 0.116) but imprecise. Appendix Table A14 reports the unrestricted Red/Blue/Yellow specification; the Yellow point estimates are positive but substantially less precise.

Within Nigeria, the contrast between math and numeracy reinforces the implementation interpretation. The math endline, tied to the pre-existing curriculum with 69% completion, shows stable small positive effects: +0.022 SD at T2 and +0.031 SD at T3. The numeracy endline, tied to the new DI curriculum with 26% completion, is more volatile: +0.042 SD at T2 and $-0.031$ SD at T3. A curriculum that is not delivered regularly gives the assignment rule little opportunity to affect learning, even when the intended contrast is clean on paper.

### 5.2 Assignment First Stage

We next test whether the assignment technology moved students in the way the model requires. Table A15 and Table A17 report baseline predictive power. Kenya is the high-signal case, with an incremental $R^2$ of 0.52. Liberia is the low-signal case, with $R^2$ between 0.03 and 0.07. Nigeria is intermediate, with $R^2$ between 0.11 and 0.22.

The classroom reallocation evidence then shows whether predictive content translated into realized grouping. Table 19 reports the harmonized dispersion first stage. Kenya is the clean first-stage success: within-class dispersion compresses substantially, so the sorting rule works mechanically. Liberia is different. Each sub-experiment also has $K = G = 2$, but within-class dispersion barely compresses because the diagnostic carries too little information, while class sizes increase from 32 to 42 students. Nigeria separates homogeneity from assignment most clearly. With $K = G = 3$, within-class baseline dispersion increases because grade mixing and implementation failures dominate ability sorting. If classroom homogeneity were the sufficient statistic, Nigeria's positive grouped point estimates would be especially hard to rationalize. The more natural interpretation is assignment-to-content under incomplete implementation, not homogeneity alone.

Kenya provides the sharpest revealed-payoff test. Table 20 and Figure 2 ask whether the students most directly affected by the rule benefit more. They do not: the mover ITT is 0.030 SD, the stayer ITT is 0.035 SD, and the treatment-by-mover interaction is essentially zero. Among movers, treatment effects do not rise with distance from the assigned cutoff. An omnibus predicted-assignment-gain index also has a near-zero interaction with treatment: 0.008 SD per one standard deviation of predicted assignment gain (SE = 0.068). Kenya therefore creates the intended reallocation without producing a positive average effect. The cross-grade assignment margin has little revealed payoff at observed delivery fidelity. Any high-input gain must therefore come from the interaction between accurate sorting and substantially higher assignment-channel delivery, not from assuming that correct placement already has a large payoff in the observed Kenya data.

## 5.3 Classroom-Production Diagnostics

The classroom-production diagnostics ask whether regrouping changed the learning environment in ways that could offset assignment gains. The Borusyak-Hull estimates are useful because scripted instruction gives content-assignment restrictions that are rarely available in tracking studies. But the estimates are diagnostic, not the paper's main causal object. They recover a composite peer/rank margin under assumptions that are most credible when treatment creates substantial peer-composition variation and scripted instruction fixes intended content by track. Appendix Table A29 reports the harmonized estimates across all three experiments.

Kenya is the most informative setting for this diagnostic because high signal quality and clean execution generate substantial treatment-induced reshuffling of classroom composition. The approximate decile specification gives $\hat{\zeta} = -0.134$ ($p = 0.02$), implying that a one-SD increase in leave-out peer mean is associated with a 0.13 SD reduction in own achievement. This adverse composite is large enough to matter, but Appendix Table A30 shows that exact control-function and recentered specifications are close to zero. The evidence therefore supports a weaker, but more credible, interpretation: classroom-production margins should enter the structural model, but the reduced-form peer/rank estimates should not be treated as a clean causal explanation for Kenya's null.

The other two settings illustrate the limits of this diagnostic. Liberia's estimate is near zero ($\hat{\zeta} = -0.048$, $p = 0.66$), but low signal quality means treatment-induced variation in peer composition is small. Nigeria's estimates switch sign across waves, from $+0.087$ ($p = 0.19$) at T2 ETE to $-0.109$ ($p = 0.28$) at T3 ETE. With realized low delivery fidelity and collapsed assignment execution, the treatment generated less structured peer-composition variation than intended. Nigeria therefore provides little leverage on the peer/rank channel.

Other classroom-production margins point in the same direction. Liberia's class sizes increase sharply, and controlling for class size does not eliminate the upper-track harm, suggesting costs beyond pure congestion, likely grade mixing and disruption of established routines. Kenya has no obvious negative delivery shock: class sizes fall slightly, grade mixing is limited, and lesson-completion point estimates are positive but imprecise. This makes Kenya a genuine production puzzle: the school creates the intended groups, but the realized assignment margin does not raise achievement. Nigeria is different. The designed three-group structure collapses in many schools, the placement test is non-binding, teachers deliver wrong-level content, and the DI numeracy curriculum is completed in only 26% of scheduled lessons. These failures mean that neither assignment execution nor assignment-channel delivery fidelity reaches the level required for complementarity to operate.

### 5.4 Mechanism Synthesis

Table 4 summarizes the three-channel decomposition, mapping each experiment to its position in $(\rho, \omega, \tau)$ space and identifying the binding margin.

Table 4: Three-Channel Decomposition and $(\rho, \omega, \tau)$ Summary

| | Liberia | Kenya | Nigeria |
|---|---|---|---|
| *Position in $(\rho, \omega, \tau)$ space* | | | |
| Signal quality ($\hat{\rho}$) | 0.055 | 0.533 | 0.121 |
| Assignment execution ($\hat{\omega}$) | 1.00 | 1.00 | 0.84 |
| Assignment-channel $\hat{\tau}$ | 0.38 | 0.50 | 0.31 |
| *Assignment and delivery evidence* | | | |
| $\Delta$ within-class SD | small | −0.328 | +0.104 |
| Delivery process moment | — | +2.4pp | 26% DI completion |
| *Residual classroom-production diagnostic* | | | |
| $\hat{\zeta}$ (BH estimate) | −0.048 | −0.134** | $\approx 0$ |
| $\Delta$ peer mean | +0.041 | +0.277 | −0.242 |
| Peer/rank contribution | −0.002 | −0.037 | +0.026 |
| *Organizational costs* | | | |
| $\Delta$ class size | +10.0 | −2.9 | −3.0 |
| Audit: groups as designed | — | — | ≤50% |
| **Overall ITT** | **−0.212** | **+0.031** | **+0.075** |
| Remainder (ITT − peer/rank) | −0.210 | +0.067 | +0.049 |
| **Main interpretation** | Weak signal | Payoff absent at observed fidelity | Execution and delivery failure |

*Notes:* Each column summarizes one experiment's position on the evidence used in the reduced-form interpretation. $\hat{\rho}$, $\hat{\omega}$, $\hat{\tau}$ are from the structural model (Section 7). Peer/rank contribution $= \hat{\zeta} \times \Delta\bar{\theta}$. Remainder = ITT − peer/rank contribution in the channel-summary calculation. Nigeria ITT is pooled across waves; peer contribution uses T3 ETE estimates. The Nigeria delivery process moment is the balanced level of DI numeracy completion in treatment and control. The Nigeria class-size row reports classroom-level means from Table A20; the student-weighted T3 channel-summary sample has a different class-size contrast.

The synthesis is consistent with the framework's qualitative predictions and shows why reduced-form evidence alone is incomplete. Liberia has faithful execution but a weak signal, so the peer/rank contribution is negligible and the negative remainder reflects organizational costs in the absence of useful assignment gains. Kenya has the strongest assignment first stage: sorting compresses dispersion and assignment is clean. Yet the achievement gain remains small and students with larger predicted assignment gains do not benefit more. Kenya therefore provides the key payoff restriction: accurate sorting at observed delivery fidelity is not enough. Nigeria is the implementation failure case. The ex ante design aimed at the missing high-input cell, but realized execution and assignment-channel delivery were both low. The experiment reveals the fragility of achieving the three inputs simultaneously rather than the payoff to having achieved them.

The reduced-form evidence answers one question: there is no hidden positive average effect across the observed implementations. It also identifies the empirical restrictions any model must respect: weak-signal harm in Liberia, successful sorting but no revealed payoff in Kenya, and failed execution and delivery in Nigeria. It does not answer the counterfactual question: what would the treatment effect be in a setting that achieves accurate sorting, near-clean assignment execution, and high assignment-channel delivery at the same time? Section 7 addresses that question with a structural model that separates the three primitives and recombines them in configurations not observed in the data.

# 6 Robustness

The robustness exercises are organized around four threats to the reduced-form interpretation: specification and inference, attrition, residual classroom-production channels, and measurement or distributional artifacts. None overturns the main empirical pattern. Liberia remains negative, Kenya remains close to zero despite a strong assignment first stage, and Nigeria remains positive but imprecise under the preferred two-group outcome specification.

## 6.1 Specification, Inference, and Attrition

Table A36 reports the Kenya and Liberia ITT estimates under alternative specifications: the preferred empirical-Bayes baseline control, a raw baseline-score control, and no baseline control. The Kenya estimate is stable under the two controlled specifications and remains statistically indistinguishable from zero under all specifications. The Liberia estimate remains negative throughout. Wild cluster bootstrap and Fisher randomization $p$-values are close to the conventional clustered-inference results, so the signs and magnitudes are not artifacts of one inference procedure.

Table A37 reports analogous robustness for Nigeria. The primary T3 ETE estimate is positive but imprecise across specifications; the wild cluster bootstrap $p$-value is 0.42 and the permutation $p$-value is 0.49. Attrition is the more important Nigeria concern. Table A38 shows that T2 endterm bounds are tightly positive, $[0.146, 0.180]$, while T3 endterm bounds are wide, $[-0.126, 0.298]$. We therefore treat T3 as the pre-specified primary endpoint but use the T2 endterm and stacked estimates as important checks on direction and magnitude.

## 6.2 Peer/Rank Diagnostics

The peer/rank evidence is used as a diagnostic margin. The Kenya estimate is negative in the approximate decile and ventile specifications, but Appendix Table A30 shows that exact control-function and recentered specifications are close to zero. The correct interpretation is therefore not that a clean adverse peer effect explains Kenya's null. Rather, the classroom-production margin is plausible enough to include in the structural accounting, but not sharp enough to carry the paper's causal argument.

Nigeria reinforces this caution. All main Nigeria results cluster at the academy level (42 academies), matching the level of randomization. This is the inference benchmark for both the preferred Red versus Blue/Yellow estimates and the unrestricted Red/Blue/Yellow appendix estimates. The Yellow-only point estimates are economically large but imprecise once academy-level clustering and the thin realized cell are respected, which is why the two-group collapse is the main specification and the three-group version is reported only for transparency.

## 6.3 Class Size and Measurement

Class-size changes do not explain the main track-position results. In Kenya and Liberia, adding realized class size leaves the upper-track estimates essentially unchanged (Table A39). In Nigeria, adding class size raises the Blue/Yellow estimate rather than attenuating it (Table A40), so the positive grouped point estimate is not mechanically driven by smaller classes.

Ceiling effects also do not explain the null or positive estimates. Kenya's ITT remains close to zero under Tobit and trimmed-OLS specifications (Table A41), and Nigeria's T3 ETE math estimate is similarly stable (Table A42). Variance decompositions provide a final check on distributional artifacts. In Kenya, treatment shifts variance from within-class to between-class components, as expected under successful sorting, without increasing aggregate score variance. In Liberia, within-class variance rises rather than falls, consistent with weak sorting and organizational disruption. In Nigeria, variance patterns are small and unstable across waves. These results support the interpretation that the observed effects are not artifacts of censoring or simple distributional reshuffling.

# 7 Structural Estimation and Counterfactuals

The reduced form leaves open the policy question that motivates the paper: what would cross-grade ability grouping produce in a setting that combines an informative diagnostic, faithful assignment execution, and high assignment-channel delivery fidelity? None of the observed experiments realizes that cell. The structural model uses the three observed cells to recover the missing one. It does not fit away null effects; it uses the pattern of null and weak effects to learn which inputs must be jointly present for assignment to level to raise achievement.

The design follows a simple experimental-structural logic. The primitives of the organizational technology are measured before treatment effects are fit: signal quality comes from control-group predictive content, assignment execution from rosters and implementation audits, and assignment-channel delivery fidelity from curriculum-specific process data. The production parameters are then estimated conditional on those primitives and required to match the realized experimental moments. This order prevents delivery fidelity from becoming an achievement residual. It also makes the counterfactual transparent: Kenya supplies a high-quality signal and a strong sorting first stage, Nigeria supplies weak execution and low assignment-channel fidelity, and Liberia supplies the consequences of faithfully acting on a weak signal.

## 7.1 Parameterization

The structural model parameterizes the production function from Section 2. Student $i$ in market $m$ has latent ability $\theta_i = \mu_{g(i),m} + \nu_i$ and observed baseline score $s_i = \theta_i + u_i$. Signal quality $\rho_{m,g} = \sigma^2_{\nu,g}/(\sigma^2_{\nu,g} + \sigma^2_{u,g})$ governs the informativeness of the diagnostic. Intended track assignment is deterministic, $k_i^* = \mathcal{A}_m(a_i; c_m)$. Realized assignment is mediated by a school-level implementation regime $R_j$ with misclassification matrix $M_{m,r}(k,\ell) = \Pr(k_i = \ell \mid k_i^* = k, R_j = r)$, summarized by assignment execution $\omega_m$.

The model targets the market-level treatment-control contrast implied by this mechanism, not the full distribution of potential outcomes. For market $m$ and primitive vector $(\rho, \omega, \tau)$, the model-predicted ATE is

$$\Delta_m(\rho,\omega,\tau) = \delta_m + \lambda\rho\omega\tau^{\alpha} + \varphi(1-\tau)\big(P_m + \omega_r R_m\big) - \chi_N \widetilde{N}_m - \chi_V V^g_m. \tag{20}$$

Here $P_m$ is the treatment-induced peer-mean shift, $R_m$ is the rank-composition proxy, $\widetilde{N}_m$ is class-size pressure measured in 25-student units, and $V_m^g$ is the change in within-class grade dispersion. The first term after $\delta_m$ is the assignment-payoff channel: accurate sorting matters when the signal is informative, intended assignments are executed, and the assigned curriculum is delivered. The second term is the social channel. It is scaled by $(1 - \tau)$, so higher assignment-channel fidelity both raises the payoff to correct assignment and narrows the scope for peer composition to affect instruction. The remaining terms capture production costs from larger or more grade-heterogeneous classes.

The mechanical sorting first stage is fit separately from the outcome payoff:

$$C_m = \sigma_{\varepsilon m} + \chi_V(1 - \omega_m) - \kappa\rho_m\omega_m, \tag{21}$$

where $C_m$ is the treatment-induced change in within-class dispersion. Kenya mechanically compresses classrooms even though achievement does not rise. The parameter $\kappa$ lets the model match that sorting compression without forcing the achievement payoff $\lambda$ to be large at Kenya's realized fidelity.

The nonlinear delivery-activation term $\tau_m^\alpha$ is anchored by the Kenya assignment-payoff tests in Table 20. Those tests show that the implemented assignment margin has near-zero revealed payoff at Kenya's realized fidelity. This is an identifying restriction, not a device for raising the counterfactual. The model cannot assume a large return to correct placement at observed $\tau$. It can generate large gains only if the full set of moments supports a production technology in which accurate sorting becomes valuable when the assigned curriculum is actually delivered.

## 7.2 Structural Design and Identification

The model is estimated in four sequential blocks. The sequencing plays the same role as a validation protocol in structural work built around experiments: the model first fixes the empirical objects observed outside the treatment-effect residual, then asks whether a common production map can reproduce the experimental cells actually observed. Formally, let $\hat{m}_b$ denote the empirical moments assigned to block $b$ and $m_b(\Theta_b)$ the corresponding model moments. For the blocks estimated by minimum distance, the estimator solves

$$\hat{\Theta}_b = \arg \min_{\Theta_b \in \mathcal{T}_b} [\hat{m}_b - m_b(\Theta_b)]' W_b [\hat{m}_b - m_b(\Theta_b)] + \mathcal{R}_b(\Theta_b), \tag{22}$$

where $W_b$ is diagonal. Experimental coefficients use inverse estimated variances; the deterministic dispersion-compression moment uses a scale-normalizing weight; and fixed primitives enter as conditioning inputs rather than as fitted treatment-effect moments. The term $\mathcal{R}_b(\Theta_b)$ contains the scale and shrinkage restrictions described below.

The first three blocks estimate the primitive inputs without using treatment-effect residuals. The signal-quality block uses only control-group predictive content: within each market and grade, $\rho_{mg}$ is the squared correlation between the baseline diagnostic and the endline control outcome, and the market-level $\hat{\rho}_m$ is the control-sample-weighted average across grades. The assignment-execution block uses treatment-arm roster and audit moments: track shares, score-group rank correlation, Yellow-track formation, missing assignments, and cutoff compliance. Kenya and Liberia are deterministic by design and by realized cutoff compliance, so $\omega = 1$ in both. Nigeria's execution parameter is estimated from canonical roster moments, with external audit targets reserved for validation when they conflict with cleaned roster moments. The delivery-fidelity block uses curriculum-specific process data only. For Kenya and Liberia, lesson-completion differences in the assignment channel are mapped into a bounded process index. For Nigeria, the index is based on DI numeracy completion in treatment and control, because this is the curriculum that carries the assignment-rule treatment; wrong-track delivery is retained as validation evidence rather than an outcome target.

The fourth block estimates the common production function conditional on $(\hat{\rho}, \hat{\omega}, \hat{\tau})$. The fitted moments are the harmonized market-specific ITTs, the peer/rank composites $\hat{\zeta}_m$, the treatment-induced change in within-class dispersion, and the Kenya assignment-payoff slope. Other reallocation quantities—peer shifts, class-size pressure, and grade dispersion—enter equation (20) as measured environment inputs rather than as additional fitted targets. This division of moments is what prevents the structural model from simply relabeling unobserved country heterogeneity as implementation.

With only three market-level experiments, the preferred estimator imposes three restrictions.

First, market residuals are shrinkage-regularized with a prior standard deviation of 0.25, so the model is not free to match country ITTs mechanically. Second, weak scale normalizations keep the assignment-payoff and social-channel coefficients in economically plausible ranges. These normalizations matter for exact magnitudes, but not for the complementarity result: with only three implementation regimes, the experiments identify the sign and nonadditivity of the production map more tightly than they identify the absolute scale of the high-fidelity extrapolation. Third, hard penalties rule out counterfactual mappings that violate the model's comparative statics, such as higher delivery fidelity reducing the payoff to Kenya's sorting rule. The robustness tables below relax these restrictions. Confidence intervals in the main counterfactual table perturb the stage-4 target moments using their estimated variances and re-estimate the production block, holding the first three primitive estimates fixed; Appendix C also reports an approximate combined uncertainty exercise that pairs primitive draws with the stage-4 production-parameter draws.

Identification of the production parameters follows from this division of moments. The dispersion equation pins down $\kappa$, because $\rho_m \omega_m$ predicts how much classroom compression the realized sorting rule should mechanically generate. The ITT moments and Kenya assignment-payoff slope jointly identify $\lambda$ and $\alpha$: Kenya has high $\rho$ and clean $\omega$, but its realized assignment-payoff slope is near zero at moderate $\tau$, so the model cannot obtain the high-input effect by simply assigning a large payoff to correct placement at observed fidelity. The peer/rank composites identify $\varphi$ and the rank weight $\omega_r$, but these are the least cleanly separated parameters; the preferred interpretation is therefore a composite social channel, and Appendix C reports a specification fixing $\omega_r = 0$. The class-size and grade-dispersion coefficients, $\chi_N$ and $\chi_V$, are cost-accounting terms identified from measured reallocation inputs and the ITT fit, not independent experimental estimates of class-size or grade-mixing effects.

Table 5 summarizes the distinction between fitted stage-4 moments and measured inputs.

Table 5: Stage-4 Moment Blocks and Inputs

| Block | Moments / inputs | Criterion weight | Role in estimation |
|---|---|---|---|
| Observed treatment effects | 3 market ITTs | inverse variance (54.9–356.0) | Fit ATEs while residuals are shrinkage-regularized. |
| Social-channel diagnostics | 3 peer/rank composites | inverse diagnostic variance (100.0–200.0) | Discipline the common peer/rank scale. |
| Sorting first stage | 3 within-class dispersion changes | scale weight (1000.0) | Separate sorting compression from assignment payoff. |
| Revealed assignment payoff | Kenya treatment × predicted assignment-gain slope | inverse variance (216.3) | Pin down delivery activation at realized fidelity. |
| Fixed primitives and environment | $(\hat{\rho}, \hat{\omega}, \hat{\tau})$; peer shifts, class size, grade dispersion | not criterion targets | Measured inputs for observed and counterfactual regimes. |

*Notes:* The stage-4 minimum-distance criterion targets only moments with a criterion weight in this table. Classroom peer shifts, class size, grade dispersion, and the first-three-block primitives enter the production mapping as measured inputs, not as additional fitted moments. This distinction prevents the target list from overstating what the outcome block estimates.

Appendix C reports the local and numerical diagnostics behind this estimator. The weighted Jacobian for the production block has numerical rank seven for the seven common production parameters, but it is ill-conditioned. This matters for interpretation: the model is well suited to the paper's counterfactual question—how the observed implementation regimes map into the missing high-input cell—but individual production parameters should not be read as sharply estimated primitive technologies. The same appendix shows that the optimum is not being held in place by binding inequality penalties, that all 25 optimizer starts converged, that the reported conditional intervals use 49 converged parametric re-estimations of the stage-4 moments, and that the high-input counterfactual remains positive in the lower tail of an approximate combined primitive-plus-production uncertainty exercise.

Table 6 reports the primitive-specific moments that pin down the first three inputs before the treatment-effect fit. The table highlights two features of the design. First, signal quality is not inferred from treatment effects: it is the control-group predictive content of the baseline diagnostic. Second, Nigeria's low execution and delivery fidelity are measured on different margins. The former comes from realized rosters and group formation; the latter comes from DI numeracy completion, the process measure for the assignment channel. This separation is what allows the counterfactuals below to distinguish a logistical failure to form groups from a pedagogical failure to deliver the content assigned to those groups.

Table 6: Primitive Identification Moments

| Primitive | Moment | Data | Model / estimate |
|---|---|---|---|
| *Block 1: Signal quality from control-group measurement moments* | | | |
| $\rho$ | Kenya: baseline–endline $R^2$ (control, $N = 4,195$) | 0.533 | $\hat{\rho} = 0.533$ |
| $\rho$ | Liberia: baseline–endline $R^2$ (control, $N = 1,482$) | 0.055 | $\hat{\rho} = 0.055$ |
| $\rho$ | Nigeria: baseline–endline $R^2$ (control, $N = 413$) | 0.121 | $\hat{\rho} = 0.121$ |
| *Block 2: Assignment execution from audit/roster moments* | | | |
| $\omega$ | Kenya: cutoff misclassification rate | 0.000 | $\hat{\omega} = 1.00$ |
| $\omega$ | Liberia: cutoff misclassification rate | 0.000 | $\hat{\omega} = 1.00$ |
| $\omega$ | Nigeria: score–group rank correlation | 0.249 | 0.253 |
| $\omega$ | Nigeria: share schools with $\leq$2 groups | 0.200 | 0.203 |
| $\omega$ | Nigeria: share schools with any Yellow track | 0.800 | 0.789 |
| $\omega$ | Nigeria: share missing assignment | 0.348 | 0.340 |
| *Block 3: Delivery fidelity from assignment-channel process moments* | | | |
| $\tau$ | Kenya: lesson-completion treatment–control difference | 0.025 | $\hat{\tau} = 0.50$ |
| $\tau$ | Liberia: lesson-completion treatment–control difference | 0.011 | $\hat{\tau} = 0.38$ |
| $\tau$ | Nigeria: DI numeracy completion, treatment | 0.262 | $\hat{\tau} = 0.31$ |
| $\tau$ | Nigeria: DI numeracy completion, control | 0.261 | $\hat{\tau} = 0.31$ |

*Notes:* The table reports the primitive-specific moments used to pin down the first three structural primitives before the treatment-effect fit. Signal quality is the market-level control-group baseline–endline $R^2$. Kenya and Liberia assignment execution is deterministic cutoff compliance; Nigeria execution is fit to the cleaned roster/audit moments shown here, with conflicting external audit targets reserved for validation. Delivery fidelity is based on lesson-completion moments in the assignment channel; Nigeria uses DI numeracy completion because that curriculum carries the assignment-rule treatment.

### 7.3 Parameter Estimates

Table 7 reports the structural estimates.

Table 7: Structural Parameter Estimates

| | Liberia | Kenya | Nigeria |
|---|---|---|---|
| *Block 1: Signal quality (control-group moments)* | | | |
| $\hat{\rho}$ | 0.055 | 0.533 | 0.121 |
| *Block 2: Assignment execution (audit/roster moments)* | | | |
| $\hat{\omega}$ | 1.00 | 1.00 | 0.84 |
| *Block 3: Assignment-channel delivery fidelity (process moments)* | | | |
| $\hat{\tau}$ | 0.38 | 0.50 | 0.31 |
| *Block 4: Production parameters (common across markets)* | | | |
| $\hat{\lambda}$ (assignment-payoff scale) | | 0.400 | |
| $\hat{\varphi}$ (social channel) | | 0.192 | |
| $\hat{\omega}_r$ (rank weight in composite) | | $-0.281$ | |
| $\hat{\chi}_N$ (class-size pressure) | | 0.033 | |
| $\hat{\chi}_V$ (grade dispersion) | | 0.022 | |
| $\hat{\alpha}$ (delivery activation) | | 4.317 | |
| $\hat{\kappa}$ (sorting compression) | | 1.649 | |

*Notes:* Blocks 1–3 measure or calibrate primitives sequentially before fitting treatment-effect moments: control predictive content for $\rho$, implementation evidence for $\omega$, and process/delivery evidence for $\tau$. Block 4 is estimated by minimum distance conditional on fixed $(\hat{\rho}, \hat{\omega}, \hat{\tau})$, targeting harmonized ITTs, peer/rank composites, within-class dispersion changes, and the Kenya assignment-payoff slope. $\lambda$, $\alpha$, $\kappa$, $\varphi$, $\omega_r$, $\chi_N$, and $\chi_V$ are constrained to be common across markets; $\delta_m$ is market-specific and shrinkage-regularized.

The estimates have three main implications. First, the $\rho$ ranking recovers the intended ordering: Kenya's diagnostic is highly informative, Liberia's is near-random, and Nigeria's is moderate. Second, $\tau$ does *not* follow the operator's ex ante general delivery ranking. The operator ranked Nigeria highest on general delivery quality (Table 1), but assignment-channel $\tau$ is lowest in Nigeria (0.31) because the DI numeracy curriculum was not absorbed into routine practice. In Kenya, by contrast, the literacy revision was delivered through the established programme, yielding the highest $\hat{\tau}$ (0.50). Nigeria therefore failed on two dimensions, not one: it did not maintain the intended groups, and it did not deliver the curriculum through which those groups were meant to matter. Third, the estimated activation parameter $\hat{\alpha} = 4.32$ is large. This is not a nuisance feature of the estimator; it is the parameter that makes the model respect Kenya. Kenya has a highly informative signal, clean execution, and a strong sorting first stage, but the realized assignment-payoff slope is essentially zero at observed fidelity. The model can fit that fact and still generate gains of the magnitude found in Duflo et al. (2011) only if assignment payoffs rise steeply when the assigned curriculum is delivered faithfully.

Appendix Tables A49 and A50 give the corresponding uncertainty and social-channel checks. The assignment-payoff scale and social-channel scale remain positive across the stage-4 re-estimation draws, and the delivery-activation parameter remains above one even at the 5th percentile. The rank weight inside the peer/rank composite is weakly identified, but fixing it to zero leaves the high-input ATE nearly unchanged (+0.187 to +0.181 SD). The structural result is therefore not a claim about a separately identified rank effect; it is a claim about assignment payoffs that become valuable when the assigned content is actually delivered.

## 7.4 Model Fit

Table 8 compares targeted moments to model-generated moments.

Table 8: Targeted Moments: Data vs. Model

| Market | Moment | Data | Model | Std. resid. | Obj. |
|---|---|---|---|---|---|
| Kenya | ITT | +0.031 | +0.030 | +0.021 | 0.000 |
| Kenya | $\hat{\zeta}$ (peer/rank) | −0.134 | −0.090 | −0.616 | 0.379 |
| Kenya | $\Delta$ within-class SD | −0.193 | −0.193 | −0.000 | 0.000 |
| Kenya | assignment-payoff slope | +0.008 | +0.011 | −0.040 | 0.002 |
| Liberia | ITT | −0.212 | −0.176 | −0.266 | 0.071 |
| Liberia | $\hat{\zeta}$ (peer/rank) | −0.048 | −0.111 | +0.633 | 0.401 |
| Liberia | $\Delta$ within-class SD | +0.011 | +0.011 | −0.003 | 0.000 |
| Nigeria | ITT | +0.093 | +0.056 | +0.281 | 0.079 |
| Nigeria | $\hat{\zeta}$ (peer/rank) | −0.109 | −0.125 | +0.155 | 0.024 |
| Nigeria | $\Delta$ within-class SD | +0.021 | +0.021 | −0.001 | 0.000 |

*Notes:* "Data" reports the reduced-form point estimate. "Model" reports the model-implied moment at the estimated parameter values. "Std. resid." is $(\text{Data}-\text{Model})\sqrt{w}$ and "Obj." is the moment's contribution to the stage-4 criterion. ITT targets are the harmonized study-specific estimates from Table 16. The assignment-payoff slope is the coefficient on treatment interacted with the standardized predicted assignment-gain index in Table 20. Because market residuals are shrinkage-regularized, the model is not forced to interpolate the three ITTs exactly.

The model fits Kenya's ITT and within-class dispersion compression closely and keeps the Kenya assignment-payoff slope near zero. It does not exactly interpolate Liberia and Nigeria. A version with unrestricted market residuals can mechanically match all three ITTs, but doing so loads too much of Nigeria's observed positive point estimate onto a nuisance intercept. The preferred specification shrinkage-regularizes those residuals, fitting Liberia and Nigeria within sampling uncertainty while preserving the common production structure. Table 8 reports standardized residuals and objective contributions. The largest weighted residual is the Liberia peer/rank moment (0.63), followed by Kenya's peer/rank moment (−0.62); the largest ITT residual is only 0.28 standard errors. This pattern is informative: the reduced-form peer/rank estimates are noisy and specification-sensitive, so the structural counterfactuals lean more heavily on the stage-specific assignment, delivery, and ITT moments than on a separately identified peer coefficient.

We also use the pooled estimates in Table 16 as validation rather than as additional targets, since they reuse the same country-level ITT information. The fitted model implies a fixed-effect meta-analytic observed effect of +0.009 SD, close to the reduced-form meta estimate of +0.010 SD. The student-weighted IPD estimate is more negative (−0.070 SD) than the model's aggregate observed-effect approximation (−0.032 SD), reflecting that the IPD estimand is a residualized common-effect regression rather than a simple average of study ATEs. The model also understates cross-study heterogeneity ($I^2 = 7.7\%$ in the model vs. 38.8% in the reduced form), the price of imposing a common production structure with shrunken market residuals. This distinction reinforces the intended use of the structural model: it estimates the missing high-$\rho$, high-$\omega$, high-$\tau$ cell rather than reproducing every reduced-form contrast exactly.

Appendix C reports the robustness battery. The main lesson is that the sign and complementarity of the high-input counterfactual are more stable than the exact magnitude. Varying residual shrinkage, perturbing the observed-$\tau$ calibration, dropping target blocks, and omitting individual markets all leave the high-input assignment-delivery counterfactual positive and economically meaningful in the preferred scale-normalized specifications. The magnitude is less invariant to auxiliary scale restrictions: removing all scale information lowers the estimate substantially, while weaker normalizations keep it close to the preferred value. The delivery-activation sensitivity explains why the preferred nonlinear payoff is not an arbitrary extrapolation. A linear activation specification fits the three ITTs about as well, but fails the Kenya revealed-payoff diagnostic by predicting a positive assignment-payoff slope where the target is near zero. Specifications that fit this diagnostic imply gains around +0.18–+0.19 SD. Thus the benchmark magnitude is tied to the observed combination of Kenya's strong sorting first stage, near-zero realized assignment payoff, and transparent scale/calibration choices, not to unrestricted interpolation.

### 7.5 Model Validation and Counterfactuals

Before using the model for the missing high-input cell, we require it to pass three sets of checks. The first is measurement validity: the primitives must line up with the data margins that identify them before treatment effects are fit. Signal quality must rank Kenya as high predictive content, Liberia as weak predictive content, and Nigeria as intermediate; assignment execution must be near-deterministic in Kenya and Liberia and lower in Nigeria; and assignment-channel delivery fidelity must come from process data rather than from achievement residuals. The second is economic validity: the production map must satisfy the comparative statics implied by the theory. Raising delivery fidelity for Kenya's sorting rule cannot lower the predicted ATE, raising execution in Nigeria cannot lower the predicted ATE, and the high-input cell must dominate the observed cells. The third is experimental fit: at observed primitive values, the model must remain close to the reduced-form treatment effects and first-stage moments generated by the experiments.

Table 9 reports these checks. They are not decorative robustness exercises. They are the acceptance criteria for interpreting the counterfactual: a model that inferred delivery from achievement, reversed the primitive rankings, violated the comparative statics, or missed the observed cells would not be used for the paper's headline result.

Table 9: Structural Validation Checks

| Check | Evidence | Status |
|---|---|---|
| Signal-quality ranking | $\rho_K = 0.533 > 0.121 > 0.055 = \rho_L$ | Pass |
| Assignment-execution ranking | $\omega_K = \omega_L = 1.00 > 0.84 = \omega_N$ | Pass |
| Delivery primitive measured before outcomes | $\tau$ calibrated from process moments before stage 4 | Pass |
| Kenya delivery monotonicity | ATE rises from +0.030 to +0.155 | Pass |
| Nigeria execution monotonicity | ATE weakly rises from +0.0561 to +0.0562 | Pass |
| Nigeria target coherence | Cleaned roster moments estimated; conflicting audit targets validation only | Pass |
| Observed-cell fit | Predicted observed ATEs: K +0.030, L −0.176, N +0.056; max ITT error/SE 0.281 | Pass |
| High-input dominance | High-input +0.187 exceeds max observed +0.056 | Pass |

*Notes:* The table reports post-estimation restrictions used before interpreting the counterfactuals. These checks are not additional moments that mechanically fit the high-input result. They verify that the measured primitives line up with their identifying evidence outside the treatment-effect fit, that the production mapping respects the model's comparative statics, and that fitted observed-cell ATEs remain close to the harmonized reduced-form estimates.

This is not a holdout validation design in the strict sense; all three country experiments enter the production map. The validation comes instead from separation of margins. The model cannot use treatment effects to choose the primitives, cannot use country residuals to interpolate the ITTs, and cannot create the high-input effect through a mapping that fails the observed-cell diagnostics. The high-input counterfactual below is therefore not obtained by assuming away the observed nulls. It is obtained by combining an observed high-quality signal, near-clean execution, and high assignment-channel delivery inside a common production map that also fits the observed experiments.

### 7.6 Counterfactuals

Table 10 reports the counterfactual ATEs, and Figure 4 displays the main comparisons. The observed panel reports all three experiments. The counterfactual panels answer two questions. First, what would the Kenya sorting rule have produced if its high signal quality and clean execution had been paired with high assignment-channel delivery? Second, which Nigeria margin was most binding: the visible collapse in group formation, the lower diagnostic content, or the failure to deliver the assigned DI numeracy curriculum? Liberia does not appear as a separate counterfactual panel because it is not a promising high-input environment; its role is to identify the cost of faithfully executing a weak signal.

Table 10: Counterfactual Treatment Effects

| Scenario | ATE | 90% interval | Description |
|---|---|---|---|
| *Observed* | | | |
| Kenya observed | +0.030 | [−0.066, +0.117] | $\rho = 0.53$, $\omega = 1.00$, $\tau = 0.50$ |
| Liberia observed | −0.176 | [−0.337, −0.006] | $\rho = 0.06$, $\omega = 1.00$, $\tau = 0.38$ |
| Nigeria realized | +0.056 | [−0.148, +0.244] | $\rho = 0.12$, $\omega = 0.84$, $\tau = 0.31$ |
| *Counterfactual 1: Kenya with higher $\tau$* | | | |
| Kenya high-$\tau$ | +0.155 | [+0.028, +0.206] | $\rho = 0.53$, $\omega = 1.00$, $\tau = 0.90$ |
| *Counterfactual 2: Nigeria as designed* | | | |
| Nigeria designed $\omega$ | +0.056 | [−0.148, +0.245] | $\rho = 0.12$, $\omega = 0.95$, $\tau = 0.31$ |
| *Counterfactual 3: Fully idealized* | | | |
| High $\rho$, high $\omega$, high $\tau$ | +0.187 | [+0.058, +0.222] | $\rho = 0.53$, $\omega = 0.98$, $\tau = 0.95$ |
| *One-at-a-time decomposition (from Nigeria baseline)* | | | |
| Upgrade $\rho$ only | +0.057 | [−0.141, +0.263] | $\rho \to 0.53$, $\omega = 0.84$, $\tau = 0.31$ |
| Upgrade $\tau$ only | +0.115 | [−0.105, +0.306] | $\rho = 0.12$, $\omega = 0.84$, $\tau = 0.90$ |
| Upgrade $\omega$ only | +0.056 | [−0.148, +0.245] | $\rho = 0.12$, $\omega = 0.95$, $\tau = 0.31$ |

*Notes:* Intervals are 5th–95th percentiles from 49 parametric bootstrap re-estimation draws of the targeted stage-4 reduced-form moments, conditional on the stage-1 to stage-3 primitive estimates. The observed panel reports all three experiments. Counterfactual rows focus on the missing policy cells anchored by the design: Kenya with higher assignment-channel delivery, Nigeria with improved execution or delivery, and the fully high-input benchmark. Liberia enters the structural fit through its observed ITT, peer/rank, dispersion, and primitive moments; it is not reported as a separate high-input counterfactual because its role in the design is the low-signal, high-execution observed cell. The high-$\tau$ benchmark is $\tau = 0.90$ for the Kenya and Nigeria one-at-a-time counterfactuals. The fully idealized case uses $\omega = 0.98$ and $\tau = 0.95$. Nigeria designed execution sets $\omega = 0.95$. The one-at-a-time decomposition upgrades each primitive separately from Nigeria's realized values.

Table 11 decomposes the model-implied ATEs into structural components for the counterfactual scenarios just reported. The table is not meant to re-display every observed country cell: Liberia's observed fit is reported in Table 10 and Table 8, while this decomposition focuses on the high-signal/high-delivery and Nigeria-design exercises.

The fully high-input counterfactual is driven by the assignment-payoff term (+0.167 SD). Setting the Kenya residual to zero while holding the common production parameters and measured environment inputs fixed leaves a predicted ATE of +0.162 SD, so the result is not a country-residual artifact. The social channel is essentially shut down when $\tau$ is high, and class/grade pressure is small in the Kenya production environment. Nigeria's realized positive point estimate, by contrast, is not an assignment story: the assignment component is nearly zero, while adverse social and class/grade terms are offset by a positive market residual. This is why the Nigeria complementarity exercise below separates the productive assignment term from the country-level residual.

Table 11: Structural Decomposition of Counterfactual ATEs

| Scenario | Assign. | Social | Class/grade | Residual | ATE excl. resid. | ATE |
|---|---|---|---|---|---|---|
| Kenya observed | +0.011 | −0.000 | −0.005 | +0.025 | +0.005 | +0.030 |
| Kenya high delivery | +0.135 | −0.000 | −0.005 | +0.025 | +0.130 | +0.155 |
| Nigeria realized | +0.000 | −0.039 | −0.040 | +0.134 | −0.078 | +0.056 |
| Nigeria high signal + delivery | +0.114 | −0.006 | −0.040 | +0.134 | +0.069 | +0.203 |
| Nigeria all three | +0.128 | −0.006 | −0.040 | +0.134 | +0.083 | +0.217 |
| Fully high-input cell | +0.167 | −0.000 | −0.005 | +0.025 | +0.162 | +0.187 |

*Notes:* Assignment is $\hat{\lambda}\rho\omega\tau^{\hat{\alpha}}$. Social is the peer/rank composite scaled by $(1-\tau)$. Class/grade combines the class-size pressure and within-class grade-dispersion terms. Residual is the shrinkage-regularized market productivity shifter $\delta_m$; it is reported to make the accounting transparent, not interpreted as a mechanism. "ATE excl. resid." sets $\delta_m = 0$ while holding all common production parameters and measured environment inputs fixed. Country coverage follows Table 10: the table decomposes the counterfactual rows and the Nigeria realized comparison used in the mechanism exercise. Liberia's observed fit is reported in Table 10 and Table 8; it is not a separate counterfactual scenario in this decomposition.

The first counterfactual holds Kenya's high $\rho$ and deterministic $\omega$ fixed, then raises delivery fidelity to the upper benchmark. The predicted ATE rises from +0.030 to +0.155 SD, with a conditional 90% interval of [+0.028, +0.206]. The mechanism is complementarity: higher $\tau$ activates the return to Kenya's accurate sorting (through $\tau^{\alpha}$) and attenuates the adverse peer/rank composite (through $(1-\tau)$). This does not mean that Kenya directly proves moderate $\tau$ is the only binding constraint. The reduced-form evidence cannot rule out residual classroom-production costs, and the relatively strong incumbent grade signal in Kenya likely limited the incremental assignment margin. Kenya instead supplies the model's key negative restriction: accurate sorting and clean execution alone do not produce gains, so the preferred high-input effect must come from raising assignment-channel delivery enough for correct assignment to matter. A design that sorted accurately and produced immediate gains would show that grouping can work; the Kenya null shows that sorting, by itself, is not the treatment.

The second counterfactual asks whether Nigeria would have produced large gains had the assignment protocol been executed as designed. Replacing collapsed implementation ($\omega = 0.84$) with near-deterministic assignment execution ($\omega = 0.95$) leaves the predicted ATE essentially unchanged (+0.056 in both scenarios). Fixing assignment execution alone barely changes the treatment effect because Nigeria's signal quality ($\rho = 0.12$) and assignment-channel delivery fidelity ($\tau = 0.31$) remain low. Near-deterministic execution of a moderate signal through a low-fidelity delivery system does not produce large gains.

From Nigeria's realized baseline, the one-at-a-time decomposition reveals the relative importance of each margin. Upgrading signal quality alone to Kenya's level raises the predicted ATE only to +0.057 SD, because better information has little value when assignment-channel delivery remains low. Upgrading delivery fidelity alone to 0.90 has the largest one-at-a-time effect, raising the ATE to +0.115 SD because it both activates assignment payoffs and attenuates the social channel. Upgrading assignment execution alone to 0.95 has the smallest effect, leaving the ATE at +0.056 SD, because there is not much to execute faithfully when signal quality is moderate and delivery fidelity is low.

The one-at-a-time decomposition understates the complementarity. Table 12 evaluates all single, pairwise, and joint upgrades in the Nigeria production environment. Upgrading $\rho$ alone or $\omega$ alone leaves the ATE almost unchanged. Upgrading $\tau$ alone raises the ATE to +0.115 SD. But upgrading $\rho$ and $\tau$ jointly raises the ATE to +0.203 SD, and upgrading all three primitives raises it to +0.217 SD. The nonadditive component is large: the joint $\rho + \tau$ gain exceeds the sum of the two separate gains by +0.087 SD, and the all-three upgrade exceeds the sum of the three separate gains by +0.102 SD.

These Nigeria-environment exercises should be read as mechanism decompositions rather than as evidence that Nigeria's imprecise positive point estimate identifies the high-input payoff. Table 11 shows why: Nigeria's realized positive estimate is partly a market-residual fact, not an assignment-payoff fact. Because every row in the complementarity table holds the Nigeria residual fixed, the gain and nonadditive columns are invariant to that residual. The residual-free levels are lower—realized Nigeria is $-0.078$ SD, $\rho + \tau$ is +0.069 SD, and all three upgrades are +0.083 SD—but the gains are unchanged. The payoff to a better signal is therefore latent when assignment-channel delivery is low; it appears once the curriculum through which correct assignment matters is actually delivered.

Table 12: Complementarity of Primitive Upgrades in Nigeria

| Scenario | $\rho$ | $\omega$ | $\tau$ | ATE | Gain | Nonadd. |
|---|---|---|---|---|---|---|
| Realized Nigeria | 0.121 | 0.84 | 0.31 | +0.056 | +0.000 | +0.000 |
| Upgrade $\rho$ only | 0.533 | 0.84 | 0.31 | +0.057 | +0.001 | +0.000 |
| Upgrade $\omega$ only | 0.121 | 0.95 | 0.31 | +0.056 | +0.000 | +0.000 |
| Upgrade $\tau$ only | 0.121 | 0.84 | 0.90 | +0.115 | +0.059 | +0.000 |
| Upgrade $\rho + \omega$ | 0.533 | 0.95 | 0.31 | +0.057 | +0.001 | +0.000 |
| Upgrade $\rho + \tau$ | 0.533 | 0.84 | 0.90 | +0.203 | +0.147 | +0.087 |
| Upgrade $\omega + \tau$ | 0.121 | 0.95 | 0.90 | +0.118 | +0.062 | +0.003 |
| Upgrade all three | 0.533 | 0.95 | 0.90 | +0.217 | +0.161 | +0.102 |

*Notes:* Each row evaluates the Nigeria production environment while upgrading the indicated primitives from Nigeria's realized values. High-signal sets $\rho$ to Kenya's estimate, high-execution sets $\omega = 0.95$, and high-delivery sets $\tau = 0.90$. "Gain" is relative to realized Nigeria, and "Nonadd." is the gain beyond the sum of the corresponding one-at-a-time gains. Because every row uses the same shrinkage-regularized Nigeria residual, the gain and nonadditive columns are invariant to that residual. Country coverage is intentionally Nigeria-specific because the table asks what the failed Nigeria high-input design would have implied if individual primitives had been fixed. Kenya supplies the high-signal benchmark and Liberia enters the structural estimation as the low-signal observed cell; neither is a row in this Nigeria-environment decomposition. The comparison is designed to show complementarity among primitives, not to replace the fully idealized high-input counterfactual in Table 10.

Table 13 reports the same complementarity as a marginal-product calculation. In the Nigeria realized environment, raising signal quality from Nigeria's estimate to Kenya's estimate is worth only +0.001 SD. Holding execution fixed but raising delivery fidelity to the high-delivery benchmark raises the same signal-quality upgrade to +0.088 SD. The marginal product of the diagnostic is therefore not a fixed property of the test; it is an equilibrium return that depends on whether the organization delivers the curriculum that makes correct assignment productive.

Table 13: Marginal Return to Signal Quality by Delivery Fidelity

| Scenario | $\omega$ | $\tau$ | Gain $\rho_N \to \rho_K$ | MP of $\rho$ | Cross-partial |
|---|---|---|---|---|---|
| Nigeria realized | 0.84 | 0.31 | +0.001 | 0.002 | 0.030 |
| Nigeria high delivery | 0.84 | 0.90 | +0.088 | 0.214 | 1.025 |
| Nigeria high delivery + execution | 0.95 | 0.90 | +0.099 | 0.241 | 1.157 |
| Kenya observed delivery | 1.00 | 0.50 | +0.008 | 0.020 | 0.173 |
| Kenya high delivery | 1.00 | 0.90 | +0.105 | 0.254 | 1.218 |
| Fully high-input | 0.98 | 0.95 | +0.129 | 0.314 | 1.428 |

*Notes:* The table evaluates the assignment-payoff term $\hat{\lambda}\rho\omega\tau^{\hat{\alpha}}$. "Gain" is the ATE increase from raising signal quality from Nigeria's estimate to Kenya's estimate, holding the row's $\omega$ and $\tau$ fixed. The last two columns report the analytic marginal product of signal quality and the signal–delivery cross-partial implied by the preferred production map. Country coverage follows the object of the calculation: rows use Nigeria and Kenya because the table varies signal quality between Nigeria's moderate estimate and Kenya's high estimate. Liberia is excluded from this marginal-product table because its role is the low-signal observed cell rather than the benchmark signal-quality upgrade.

Figure 3 visualizes the same economics over the estimated response surface. Holding assignment execution near-clean, the predicted ATE is relatively flat at low delivery fidelity even as signal quality rises. The surface steepens only when high signal quality is paired with high assignment-channel delivery fidelity. The figure makes clear why the high-input result is not a one-dimensional "better sorting" result: the return to sorting is activated by delivery.

Table 14 reports the corresponding delivery-fidelity thresholds. These thresholds are an operational implication of the model, not a robustness caveat. In the Kenya production environment with high signal quality and near-clean execution, the residual-free ATE turns positive only once $\tau$ reaches about 0.43, reaches +0.10 SD at $\tau \simeq 0.85$, and reaches +0.18 SD only near $\tau \simeq 0.97$. Including the shrinkage-regularized Kenya residual shifts the thresholds down slightly, with the +0.18 SD threshold at $\tau \simeq 0.94$. Thus the model does not imply that modest delivery improvements would have transformed the observed experiments. It implies that accurate sorting becomes valuable only when assignment-channel delivery is very high.

Table 14: Delivery-Fidelity Thresholds Under High Signal and Execution

| Production env. | Residual | $\tau_{\geq 0}$ | $\tau_{\geq .10}$ | $\tau_{\geq .18}$ | ATE at $\tau = .95$ |
|---|---|---|---|---|---|
| Kenya | With $\delta_m$ | 0.00 | 0.80 | 0.94 | +0.187 |
| Kenya | $\delta_m = 0$ | 0.43 | 0.85 | 0.97 | +0.162 |
| Nigeria | With $\delta_m$ | 0.00 | 0.62 | 0.84 | +0.254 |
| Nigeria | $\delta_m = 0$ | 0.74 | 0.92 | | +0.120 |

*Notes:* Each row holds signal quality at Kenya's estimate and assignment execution near clean, then varies delivery fidelity $\tau$ in the indicated production environment. The residual-included rows use the shrinkage-regularized market residual; the residual-excluded rows set $\delta_m = 0$. Blank entries mean the target ATE is not reached for any $\tau \in [0, 1]$. Country coverage is limited to Kenya and Nigeria because the threshold exercise asks how much delivery fidelity is needed in the high-signal benchmark environment and in the failed high-input Nigeria environment. Liberia enters as the low-signal observed cell, not as a high-signal threshold environment.

This decomposition changes the reading of Nigeria. The reduced-form analysis naturally emphasizes the most visible failure—assignment execution collapse—because the audit evidence is dramatic. The model implies that even near-clean execution would not have produced large gains given Nigeria's low assignment-channel $\tau$ and moderate $\rho$. The binding margin is delivery fidelity, but delivery fidelity is most valuable when paired with accurate sorting. This is the structural sense in which the three inputs are complements rather than separable implementation fixes.

The magnitude is economically interpretable in light of Duflo et al. (2011). In the preferred specification, the fully high-input counterfactual produces an ATE of +0.187 SD, with a conditional 90% interval of $[+0.058, +0.222]$. The approximate combined primitive-plus-production interval is wider but similar in its lower tail, $[+0.057, +0.226]$. The point estimate lies in the 0.14–0.18 SD range of benchmark gains in Duflo et al. (2011), depending on specification and horizon. The interpretation is not that the observed experiments replicate DDK; they do not. DDK is within-grade tracking under teacher discretion, while our setting is cross-grade assignment to scripted content. Rather, the model reconciles the null observed implementations with gains of that order by showing that the productive inputs must be jointly present. In a teacher-discretion environment, instructional adaptation may arise endogenously from more homogeneous classrooms. In the scripted environment studied here, the analogous gain comes from accurate assignment to the correct content and faithful delivery of that content.

### 7.7 Scope and Interpretation

The structural exercise gives the paper's economic interpretation of the three experiments. The reduced form shows that the observed implementations did not generate clearly positive average effects. The model explains why: the experiments do not test a world in which the required inputs are jointly present. Liberia identifies the cost of acting on a weak signal, Kenya shows that accurate sorting and clean execution are not enough when delivery fidelity is moderate and the incumbent grade rule is already relatively informative, and Nigeria reveals the consequences of failing to maintain groups and deliver the curriculum carrying the assignment-rule contrast. The model then asks the policy counterfactual that no single experiment realizes: what would the same assignment technology produce with an informative placement signal, near-clean execution, and high delivery fidelity? The preferred benchmark is approximately +0.19 SD, in the range of the gains found by Duflo et al. (2011) under teacher discretion.

This interpretation rests on three restrictions. First, the key inputs are not chosen to fit the treatment effects. Signal quality, assignment execution, and delivery fidelity are measured or calibrated from distinct data margins before the production block is estimated. Second, the production map is constrained to obey the economics of the model: higher delivery must activate the payoff to correct assignment, and better execution has high value only when there is useful information and delivered content to execute. Third, the model must still fit the observed ITTs and first-stage moments at the realized country inputs. Kenya's revealed-payoff diagnostic is therefore not an auxiliary fact that the model explains away; it is one of the moments that prevents the high-input benchmark from being a mechanical extrapolation from sorting accuracy. The high-input result is the implication of combining the best observed signal with organizational inputs that the observed experiments failed to jointly deliver, while also matching the fact that Kenya's accurate sorting had little payoff at observed delivery.

The main claim is complementarity, not an exact point estimate. Across the robustness exercises, the sign of the high-input counterfactual and the nonadditivity between signal quality and delivery fidelity are stable. The exact estimate depends on scale normalizations and on the calibrated mapping from process data into delivery fidelity, as any counterfactual of this kind must. But the economic message does not depend on treating 0.187 as a sufficient statistic. The model-implied effect is positive and economically meaningful in the high-input regime, and one-at-a-time implementation fixes have low returns when the complementary inputs are absent.

Peer composition, rank, stigma, and teacher expectations may matter in some tracking environments, but the reduced-form peer/rank diagnostics here are not sharp enough to estimate a separate social production technology. That limitation is consistent with the design's purpose. In this setting, scripted instruction makes assignment to content the main mechanism, and the structural exercise is an assignment-delivery counterfactual. Residual peer and rank channels are allowed to shape the fit, but they are not the object on which the paper's main result rests.

The same reasoning explains the treatment of cross-country residuals. Market residuals are regularized rather than used to force a perfect country-by-country fit, because the goal is to learn about the missing high-input cell rather than to relabel all residual heterogeneity as structure. This choice is conservative for the headline exercise: the preferred high-input result is not driven by a favorable country residual, and the residual-free component remains positive. If anything, the Nigeria implementation measures may understate the full economic cost of wrong-track delivery and missing groups, which would make the return to fixing execution appear smaller than it would under a more complete measure of mis-delivery.

The right reading is affirmative but bounded. The model should not be used to rank every possible tracking design or to claim that peer effects are absent in all ability-grouping settings. It should be used for the question this paper is built to answer: whether the null realized effects in these three scripted cross-grade experiments imply that assignment to level has little value. Under the maintained structural interpretation, they do not. The evidence points to a more informative conclusion: the observed implementations lacked at least one necessary input, while the model-implied high-input benchmark is positive and economically meaningful when measurement, execution, and delivery are jointly strong.

## 8 Discussion and Conclusion

We began with a question that has divided education researchers for decades: does ability grouping help or harm students? The answer from these three experiments is not a simple verdict. None of the observed implementations produced a clearly positive average effect. But the weak effects come from different sources, and the pattern is economically informative. In Liberia, the diagnostic signal was so weak that the intervention replaced the incumbent grade signal with an inferior assignment rule. In Kenya, the diagnostic signal was strong and assignment was clean, but grade itself was also a more informative incumbent signal in a tightly managed Bridge/NewGlobe system, and accurate sorting did not translate into achievement gains at observed delivery fidelity. In Nigeria, the high-input design broke down before the assignment mechanism could be tested: group execution collapsed and the curriculum carrying the assignment-rule contrast was delivered in only 26% of scheduled lessons.

The structural model turns this pattern into the paper's main counterfactual. Under Kenya-level diagnostic quality, near-clean assignment execution, and high assignment-channel delivery fidelity, the preferred predicted ATE is about +0.19 SD. The point estimate should not be overread; the complementarity should. Improving assignment execution alone does little when delivery fidelity is low. Improving the diagnostic alone does little when the curriculum through which the diagnostic matters is not delivered. The payoff to better sorting appears only when students are assigned to the right content and that content is taught. The structural result is therefore not an adjustment to disappointing RCTs. It is the mechanism: realized effects are low because each setting is missing at least one necessary input, not because the latent value of assignment to level is zero.

This distinction helps reconcile the apparently contradictory tracking literature. The equity tradition, associated with Oakes (1985) and Gamoran (1992), documents serious harms from tracking: stigma, lower teacher expectations, weaker instructional quality in lower tracks, and widening inequality. The efficiency tradition, including the Joplin Plan evidence and the Kenya tracking experiment in Duflo et al. (2011), documents gains from grouping students so instruction can be better targeted. Both bodies of evidence are credible in their own settings. The mistake is to treat realized effects from a particular implementation regime as evidence about the latent value of ability grouping as an organizational principle.

The harms in the equity literature are not imaginary. They are implementation channels. Tracking is most likely to harm lower-track students when the signal is noisy or biased, assignment is permanent, lower tracks receive weaker teachers or less ambitious curricula, and track labels change expectations. In that regime, a negative realized effect is exactly what one should expect. But these features are not mechanical consequences of assigning students to level. They are failures of measurement, reassessment, instructional quality, and organizational design. Conversely, positive effects in the efficiency literature arise in regimes where at least some productive inputs are present: the signal has predictive power, groups are actually formed, and teachers adapt content or pace to the regrouped classroom.

Scripted instruction has an ambiguous role in this debate. It can make misassignment more costly because the teacher has less discretion to compensate for a poor placement. It may also constrain some of the equity harms emphasized by critics of tracking, although our data do not directly measure stigma, teacher expectations, or student identity. If lower-track instruction is scripted, aligned to foundational skills, and delivered with high fidelity, the label need not imply lower-quality instruction. If groups are frequently reassessed, the assignment need not be permanent. If content is determined by the lesson plan rather than by teacher beliefs about who belongs in the room, expectation effects may have less scope to shape instruction. Scripted grouping is therefore not intrinsically more or less equitable than teacher-discretion grouping. Its equity consequences depend on the same primitives as its efficiency consequences: the quality of the signal, the fidelity of execution, and the fidelity and equality of delivery across tracks.

The baseline is also not psychologically neutral. Much of the anti-tracking literature compares the stigma of being placed in a lower group to an ideal heterogeneous classroom. But a grade-based classroom in a setting with large skill dispersion is not an ideal heterogeneous classroom. It is a classroom in which many students experience daily instructional mismatch. A child who cannot read the text or follow the mathematics lesson is not spared harm because the classroom is nominally untracked. She is being sorted by grade into content she cannot learn from. The welfare comparison must weigh the psychological cost of a track label against the psychological and human-capital cost of persistent failure and invisibility in grade-level instruction. Dynamic skill complementarities make that cost especially high: missing foundations today lowers the productivity of tomorrow's instruction (Cunha and Heckman, 2007).

The broader point of the signal-substitution frame is that the policy choice is not between sorting and no sorting. It is between assignment rules. Grade is cheap and often useful, but it is a noisy proxy for readiness. A diagnostic can improve on it only if the diagnostic is informative for the instructional targets at stake and sufficiently better than the grade rule it replaces. In Liberia, it was not. Faithful execution of a bad signal was harmful. In Kenya, the diagnostic was strong, but the grade signal was also comparatively strong and delivery did not activate the remaining assignment payoff. In Nigeria, the design tried to reach the high-input cell but the organization did not execute the grouping protocol or deliver the assigned content. These are not three disconnected results. They are three revealed constraints on the same assignment technology.

For policy, the framework changes the question. Asking whether ability grouping works is too crude. A school system considering cross-grade grouping should ask four prior questions. First, how informative is the incumbent grade signal? Second, is the placement diagnostic better than the grade signal it replaces? Third, can the school form and maintain the intended groups? Fourth, will each track receive high-quality content at the intended level? If the answer to any one question is no, the intervention is not ready to deliver its latent payoff. The binding margin may be measurement, management, or instruction, and the returns to improving one margin depend on the others.

The broader implication for labor and organizational economics is that assignment policies do not operate through measurement alone. A better signal is valuable only when the organization can execute the assignment rule and when the assigned production environment actually delivers a different and productive treatment. The lesson from these experiments is that measurement, execution, and delivery are complements. Investing in better diagnostics without organizational capacity is a predictable way to learn more about people while doing little for them.

The Joplin Plan's old promise was simple: teach children where they are, not where their grade says they should be. The three experiments in this paper show why that promise is powerful and why it is difficult. Grade-based assignment is often a poor match for actual skill, so the potential gain from a better assignment rule is real. But the organizational requirements are demanding. The school must measure the right thing, act on the measurement, and deliver instruction that makes the measurement matter. None of the three observed implementations achieved all three. The structural model suggests the payoff could be large under high-fidelity implementation, but the experiments do not directly observe that regime. Whether school systems can achieve that combination at scale is the question for the next generation of work on ability grouping and targeted instruction.

## References


Angrist, N. and R. Meager (2023): "Implementation Matters: Generalizing Treatment Effects in Education," Tech. Rep. BSG-WP-2023/054, Blavatnik School of Government, University of Oxford.

Antecol, H., O. Eren, and S. Ozbeklik (2016): "Peer effects in disadvantaged primary schools evidence from a randomized experiment," *Journal of Human Resources*, 51, 95–132.

Banerjee, A., R. Banerji, J. Berry, E. Duflo, H. Kannan, S. Mukerji, M. Shotland, and M. Walton (2017): "From proof of concept to scalable policies: Challenges and solutions, with an application," *Journal of Economic Perspectives*, 31, 73–102.

Banerjee, A., S. Cole, E. Duflo, and L. Linden (2007): "Remedying education: Evidence from two randomized experiments in India," *The Quarterly Journal of Economics*, 122.

Ben-Porath, Y. (1967): "The Production of Human Capital and the Life Cycle of Earnings," *Journal of Political Economy*, 75, 352–365.

Betts, E. A. (1946): *Foundations of Reading Instruction: With Emphasis on Differentiated Guidance*, New York: American Book Company.

Borusyak, K. and P. Hull (2023): "Nonrandom exposure to exogenous shocks," *Econometrica*, 91, 2155–2185.

Burke, M. A. and T. R. Sass (2013): "Classroom peer effects and student achievement," *Journal of Labor Economics*, 31, 51–82.

Busso, M. and V. Frisancho (2023): "Ability Grouping and Student Performance: Experimental Evidence from Middle Schools in Mexico," Tech. Rep. IDB-WP-1434, Inter-American Development Bank.

Cameron, A. C., J. B. Gelbach, and D. L. Miller (2008): "Bootstrap-Based Improvements for Inference with Clustered Errors," *Review of Economics and Statistics*, 90, 414–427.

Carrell, S. E., B. I. Sacerdote, and J. E. West (2013): "From Natural Variation to Optimal Policy? The Importance of Endogenous Peer Group Formation," *Econometrica*, 81, 855–882.

Cheung, A. C. K., C. Xie, T. Zhuang, A. J. Neitzel, and R. E. Slavin (2021): "Success for All: A Quantitative Synthesis of U.S. Evaluations," *Journal of Research on Educational Effectiveness*, 14, 90–115.

Cronbach, L. J. and P. E. Meehl (1955): "Construct Validity in Psychological Tests," *Psychological Bulletin*, 52, 281–302.

Cunha, F. and J. Heckman (2007): "The Technology of Skill Formation," *American Economic Review*, 97, 31–47.

Datnow, A. and M. Castellano (2000): "Teachers' Responses to Success for All: How Beliefs, Experiences, and Adaptations Shape Implementation," *American Educational Research Journal*, 37, 775–799.

de Roux, N. and E. Riehl (2022): "Do college students benefit from placement into higher-achieving classes?" *Journal of Public Economics*, 210, 104669.

Duflo, A., J. Kiessel, and A. Lucas (2020): "Experimental Evidence on Alternative Policies to Increase Learning at Scale," Tech. rep., National Bureau of Economic Research.

Duflo, E., P. Dupas, and M. Kremer (2011): "Peer effects, teacher incentives, and the impact of tracking: Evidence from a randomized evaluation in Kenya," *American economic review*, 101, 1739–1774.

Floyd, C. (1954): "Meeting Children's Reading Needs in the Middle Grades: A Preliminary Report," *The Elementary School Journal*, 55, 99–103.

Fu, C. and N. Mehta (2018): "Ability tracking, school and parental effort, and student achievement: A structural model and estimation," *Journal of Labor Economics*, 36, 923–979.

Gamoran, A. (1992): "Is Ability Grouping Equitable?" *Educational Leadership*, 50, 11–17.

Gamoran, A. and M. Nystrand (1994): "Tracking, Instruction and Achievement," *International Journal of Educational Research*, 21, 217–231.

Garlick, R. (2018): "Academic peer effects with different group assignment policies: Residential tracking versus random assignment," *American Economic Journal: Applied Economics*, 10, 345–69.

Gutiérrez, R. and R. E. Slavin (1992): "Achievement Effects of the Nongraded Elementary School: A Best Evidence Synthesis," *Review of Educational Research*, 62, 333–376.

Hoxby, C. M. and G. Weingarth (2005): "Taking race out of the equation: School reassignment and the structure of peer effects," Tech. rep., Citeseer.

Kulik, J. A. and C.-L. C. Kulik (1987): "Effects of ability grouping on student achievement," *Equity and Excellence in Education*, 23, 22–30.

Lavrijsen, J. and I. Nicaise (2015): "New empirical evidence on the effect of educational tracking on social inequalities in reading achievement," *European Educational Research Journal*, 14, 206–221.

Lazear, E. P. (2001): "Educational Production," *Quarterly Journal of Economics*, 116, 777–803.

Lord, F. M. and M. R. Novick (1968): *Statistical Theories of Mental Test Scores*, Reading, MA: Addison-Wesley.

Muralidharan, K., A. Singh, and A. J. Ganimian (2019): "Disrupting education? Experimental evidence on technology-aided instruction in India," *American Economic Review*, 109, 1426–1460.

Oakes, J. (1985): *Keeping Track: How Schools Structure Inequality*, New Haven, CT: Yale University Press.

Rjosk, C. (2022): "Dispersion of Student Achievement and Classroom Composition," *International Handbook of Comparative Large-Scale Studies in Education: Perspectives, Methods and Findings*, 1–33.

Slavin, R. E. (1987): "Ability grouping and student achievement in elementary schools: A best-evidence synthesis," *Review of Educational Research*, 57, 293–336.

——— (1990): "Achievement effects of ability grouping in secondary schools: A best-evidence synthesis," *Review of educational research*, 60, 471–499.

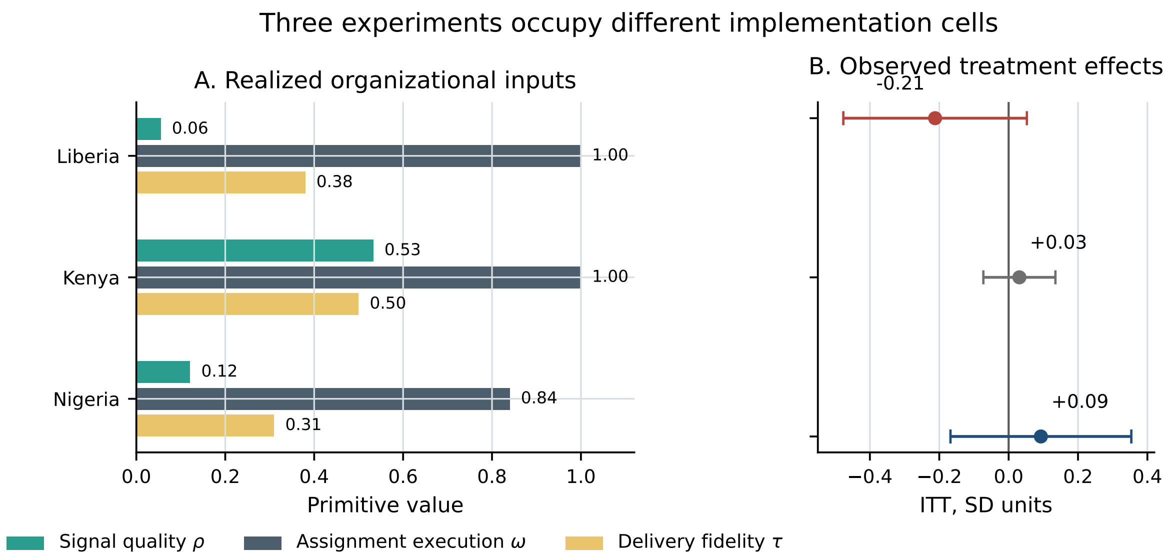
Terrin, E. and M. Triventi (2022): "The effect of school tracking on student achievement and inequality: A meta-analysis," *Review of Educational Research*, 00346543221100850.

Vigdor, J. and T. Nechyba (2007): "Peer effects in North Carolina public schools," *Schools and the equal opportunity problem*, 73–101.

Vygotsky, L. S. (1978): *Mind in Society: The Development of Higher Psychological Processes*, Cambridge, MA: Harvard University Press, edited by Michael Cole, Vera John-Steiner, Sylvia Scribner, and Ellen Souberman.

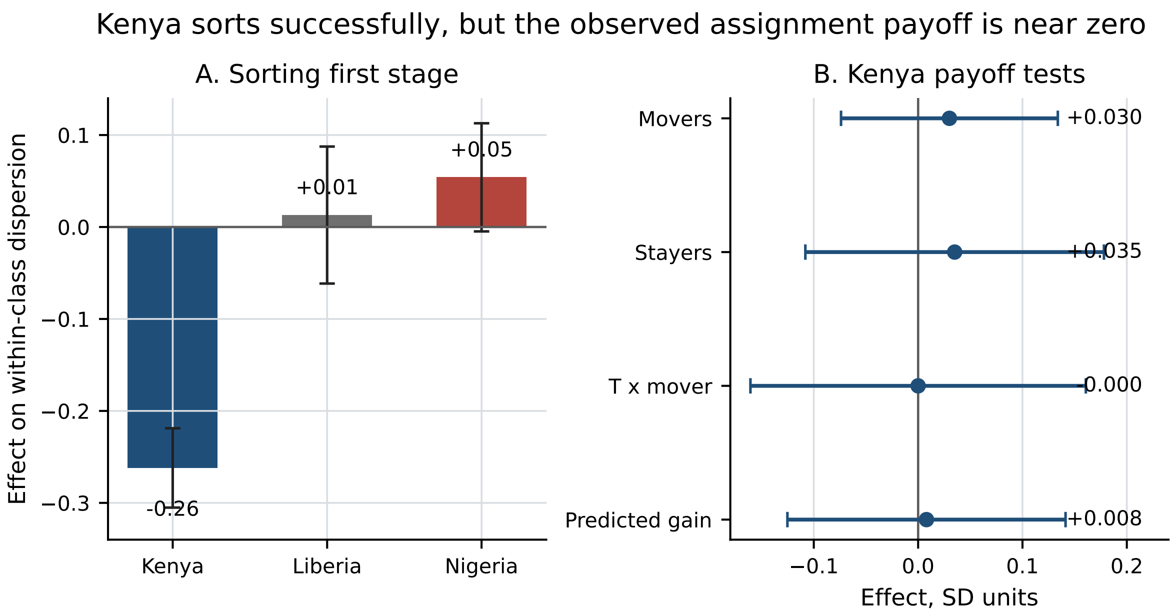
# Figures

Figure 1: Three Experiments in Implementation Space



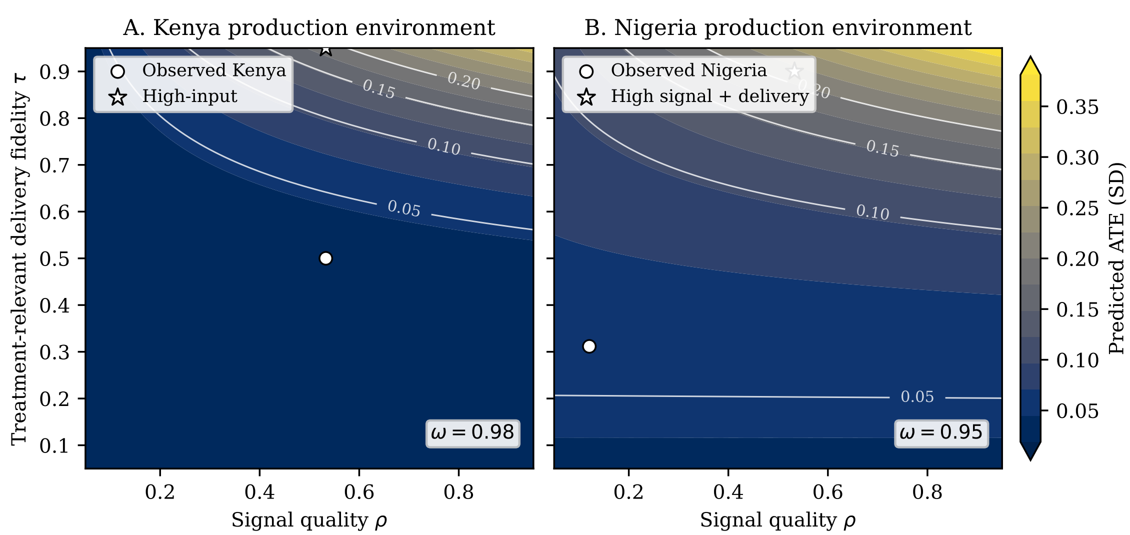

*Notes:* Panel A reports the three primitives used in the structural model: signal quality ($\rho$), assignment execution ($\omega$), and assignment-channel delivery fidelity ($\tau$). Panel B reports the corresponding reduced-form ITT estimates with 95% confidence intervals from the primary endpoint in each experiment. Values are taken from Tables 15 and 7.

Figure 2: First-Stage Sorting and Revealed Assignment Payoffs



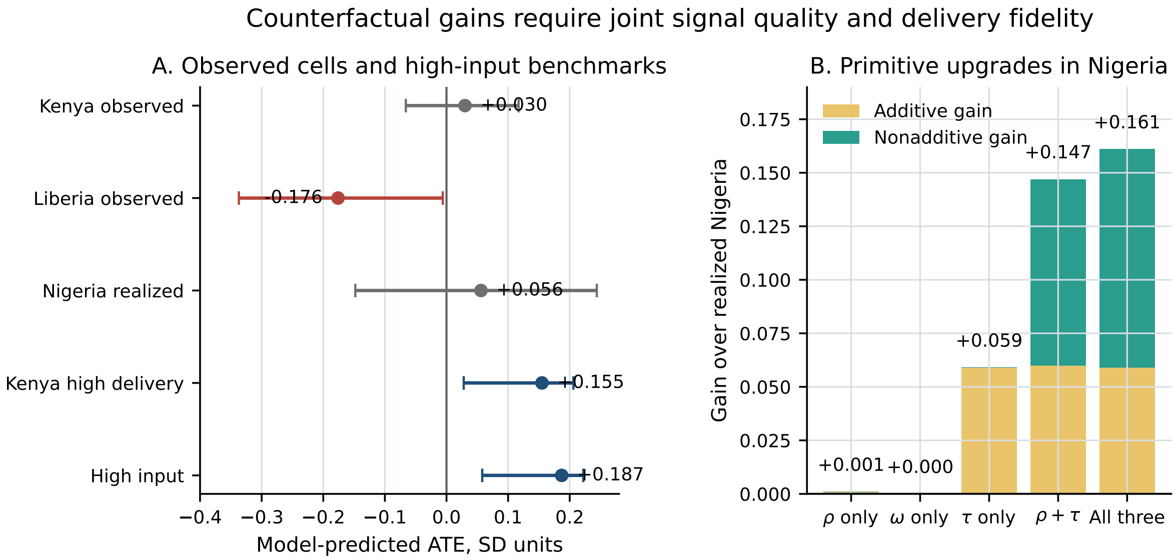

*Notes:* Panel A reports the treatment effect on within-class baseline-score dispersion from Table 19; negative values indicate greater within-class homogeneity. Panel B reports Kenya assignment-payoff diagnostics from Table 20. Error bars are 95% confidence intervals.

Figure 3: Structural Complementarity Between Signal Quality and Delivery Fidelity




*Notes:* The figure plots model-predicted ATEs over signal quality ($\rho$) and assignment-channel delivery fidelity ($\tau$), holding assignment execution near-clean. Panel A uses the Kenya production environment with $\omega = 0.98$; Panel B uses the Nigeria production environment with $\omega = 0.95$. Markers show observed $(\rho, \tau)$ values and the high-input counterfactual point. Contours report predicted ATEs in standard-deviation units.

Figure 4: Structural Counterfactuals and Complementarity




*Notes:* Panel A reports model-predicted ATEs and 90% intervals from Table 10. Panel B reports gains over realized Nigeria from Table 12. The nonadditive component is the gain beyond the sum of the corresponding one-at-a-time primitive upgrades.

## Main Tables

The main tables report the experimental estimates and first-stage diagnostics that carry the paper's argument. Pooled or harmonized estimates are included only when they answer a reader-facing question about the observed experiments. More assumption-sensitive peer/rank diagnostics and design-specific robustness checks are reported in the appendix. For Nigeria grouped effects, all main and harmonized tables use the support-aware two-track collapse, Red versus Blue/Yellow; the appendix reports one unrestricted Red/Blue/Yellow table for transparency.

### Treatment Effects

Table 15: Pooled ITT Across Kenya, Liberia, and Nigeria

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | Kenya | Liberia | Nigeria | Pooled |
| Treatment | 0.031 | -0.212 | 0.093 | -0.070 |
| | ( 0.053) | ( 0.135) | ( 0.133) | ( 0.071) |
| Control Mean | 0.012 | 0.091 | 0.036 | 0.033 |
| N | 4954 | 3154 | 1661 | 9769 |
| Pooled ITT (Nigeria T2 ETE alt) | | | | -0.051 |
| | | | | ( 0.067) |

Notes: Outcome is standardized endline score for Kenya/Liberia and T3 ETE math+numeracy index for Nigeria. Controls: country-specific baseline slopes and country-nested strata FE. SEs clustered at country-specific cluster units (school for Kenya, school-grade-group for Liberia, academy for Nigeria). Final row shows pooled robustness replacing Nigeria T3 ETE with T2 ETE index. ***, **, and * indicate 1%, 5%, and 10% significance.

Table 16: Pooled and Meta-Analytic Treatment Effects Across Experiments

| Estimand or diagnostic | Estimate | SE / statistic |
|---|---|---|
| *Common-effect estimands* | | |
| Student-weighted IPD | -0.070 | ( 0.071) |
| Experiment-balanced IPD | -0.058 | ( 0.077) |
| Fixed-effect meta-analysis | 0.010 | ( 0.046) |
| Random-effects meta-analysis | -0.008 | ( 0.078) |
| *Study-specific effects from stacked IPD* | | |
| Kenya | 0.031 | ( 0.053) |
| Liberia | -0.212 | ( 0.135) |
| Nigeria | 0.093 | ( 0.131) |
| *Heterogeneity and primitive-score diagnostics* | | |
| Treatment-effect heterogeneity test | | p = 0.196 |
| Meta-analysis Q statistic | | Q = 3.270, p = 0.195 |
| I-squared / tau-squared | | 38.8% / 0.007 |
| Treatment $\times$ ($\rho\omega\tau$) slope | 0.683 | ( 0.494) |
| Implied effect: lowest/highest score | -0.132 / 0.035 | descriptive |
| Student observations / studies | 9769 | 3 |

Notes: IPD = individual participant data. Student-weighted IPD weights each student equally; experiment-balanced IPD weights each experiment equally. Meta-analytic rows pool the three study-specific estimates, with the random-effects SE using a Hartung-Knapp-style small-sample adjustment. All IPD specifications include country-specific baseline slopes and country-nested strata FE, with SEs clustered at harmonized assignment clusters. The primitive-score slope is descriptive because $\rho\omega\tau$ varies only at the experiment level.

Table 17: Treatment Effects by Track Position Across Countries

| | (1) | (2) | (3) |
|---|---|---|---|
| | Kenya | Liberia | Nigeria |
| Lower-track effect | 0.046 | -0.019 | 0.007 |
| | ( 0.137) | ( 0.181) | ( 0.152) |
| Higher-track differential | -0.023 | -0.328 | 0.239** |
| | ( 0.147) | ( 0.201) | ( 0.098) |
| Higher-track effect | 0.023 | -0.348** | 0.246* |
| | ( 0.044) | ( 0.154) | ( 0.129) |
| N | 4954 | 3154 | 1611 |

Notes: Each column estimates a country-specific treatment-by-track-position interaction. The higher track is the upper reading group in Kenya and Liberia; in Nigeria it is the preferred two-group collapse, Blue/Yellow (`group_id` = 2 or 3). Controls are country-specific baseline slopes and country-nested strata fixed effects. SEs are clustered at the harmonized assignment cluster.

Table 18: Nigeria Reduced-Form Effects Under the Preferred Two-Group Collapse

| | T2 MTE | T2 ETE | T3 MTE | T3 ETE | Stacked ETE |
|---|---|---|---|---|---|
| *Effective two-track collapse: Red vs Blue/Yellow* | | | | | |
| Red effect | -0.035 | 0.183* | 0.159 | 0.043 | 0.138 |
| | ( 0.139) | ( 0.098) | ( 0.147) | ( 0.155) | ( 0.102) |
| Blue/Yellow effect | 0.015 | 0.173* | 0.048 | 0.236* | 0.214** |
| | ( 0.089) | ( 0.101) | ( 0.168) | ( 0.132) | ( 0.095) |
| Blue/Yellow − Red | 0.050 | -0.010 | -0.111 | 0.194 | 0.076 |
| | ( 0.104) | ( 0.103) | ( 0.201) | ( 0.116) | ( 0.080) |
| N | 1804 | 1947 | 1421 | 1611 | 3558 |

Notes: Outcomes are standardized math+numeracy index outcomes. Each column estimates a treatment-by-two-group interaction with baseline score, constituency fixed effects, and grade fixed effects; the stacked endterm column pools T2 and T3 endterm observations and adds wave fixed effects. Standard errors are clustered at the academy level. The table uses the preferred support-aware Nigeria collapse, combining Blue and Yellow into a higher-level group because the realized Yellow cell is thin. ***, **, and * indicate 1%, 5%, and 10% significance.

## Assignment First Stage and Payoff Diagnostics

Table 19: Pooled First Stage: Effect on Within-Class Dispersion

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | Kenya | Liberia | Nigeria | Pooled |
| Treatment | -0.262*** | 0.013 | 0.054* | -0.071*** |
| | ( 0.022) | ( 0.038) | ( 0.030) | ( 0.023) |
| N | 4954 | 3154 | 2354 | 10462 |

Notes: Harmonized dispersion measure is absolute deviation from classroom mean baseline ability. Nigeria uses `dev_bl`; Kenya/Liberia use `dev_eb`. Controls are country-specific baseline slopes and country-nested strata fixed effects. SEs are clustered at the harmonized assignment cluster.

Table 20: Kenya: Assignment Payoff Diagnostics

| Measure | Coef. | SE | N |
|---|---|---|---|
| *A. Randomized reallocation contrasts* | | | |
| Movers ITT | 0.030 | ( 0.053) | 1816 |
| Stayers ITT | 0.035 | ( 0.073) | 3138 |
| Treatment × mover | -0.000 | ( 0.082) | 4954 |
| *B. Assignment intensity among movers* | | | |
| G1 movers up: Treatment × cutoff distance | 0.005 | ( 0.017) | 1286 |
| G2 movers down: Treatment × cutoff distance | -0.006 | ( 0.023) | 530 |
| *C. Predicted assignment gain* | | | |
| Treatment × predicted assignment gain (1 SD) | 0.008 | ( 0.068) | 4954 |
| *D. Descriptive control-group revealed-value checks* | | | |
| Mismatch proxy | -0.008 | ( 0.011) | 4195 |
| Mismatch proxy + school-grade mean BL | -0.002 | ( 0.011) | 4195 |

*Notes:* Outcome is the standardized endline test score. Movers are students whose treatment reading-group assignment differs from their enrolled grade: Grade 1 students above the upper-track cutoff and Grade 2 students at or below the lower-track cutoff. Cutoff distance is signed so larger values indicate more clearly assigned movers. Predicted assignment gain is the standardized reduction in squared EB-score mismatch from replacing the grade target with the treatment-track target; the regression also controls for its square. Panel D is estimated only in control classrooms and is descriptive rather than randomized. Standard errors are clustered at the school level.

# A Additional Reduced-Form Tables

These appendix tables report design-specific details, sample diagnostics, and robustness checks. Kenya and Liberia appear together when the common 2018 two-track literacy design supports a shared table. Nigeria appears in adjacent tables when its three-track numeracy design or wave structure requires a separate estimand. Except for Table A14, Nigeria grouped-effect tables use the preferred two-group collapse, Red versus Blue/Yellow.

## A.1 Data and Sample

Table A1: Summary Statistics — Liberia

| | Experiment (Liberia) | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Variable | Control Mean | Treatment Mean | SD (All) | N |
| Baseline score | 21.014 | 19.848 | 7.936 | 4784 |
| Endline score | 25.983 | 24.467 | 7.612 | 3154 |
| Upper-track assignment | 0.612 | 0.549 | 0.494 | 4784 |
| Has endline | 0.659 | 0.660 | 0.474 | 4784 |
| Class size | 45.596 | 59.278 | 27.288 | 4784 |

Notes: Means and standard deviations are reported in the analytic sample. Endline outcomes are based on observations with non-missing endline test scores.

Table A2: Summary Statistics — Kenya

| | Experiment (Kenya) | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Variable | Control Mean | Treatment Mean | SD (All) | N |
| Baseline score | 39.207 | 39.904 | 8.786 | 7102 |
| Endline score | 39.333 | 40.258 | 8.519 | 4954 |
| Upper-track assignment | 0.614 | 0.652 | 0.486 | 7102 |
| Has endline | 0.686 | 0.771 | 0.459 | 7102 |
| Class size | 24.307 | 20.935 | 11.233 | 7102 |

Notes: Means and standard deviations are reported in the analytic sample. Endline outcomes are based on observations with non-missing endline test scores.

Table A3: Nigeria Summary Statistics

| Variable | Control Mean | Treatment Mean | SD (All) | N |
|---|---|---|---|---|
| Baseline score | 82.608 | 80.171 | 17.719 | 2354 |
| Observed T2 midterm | 0.634 | 0.622 | 0.483 | 3737 |
| Observed T2 endterm | 0.677 | 0.691 | 0.465 | 3737 |
| Observed T3 midterm | 0.516 | 0.470 | 0.500 | 3737 |
| Observed T3 endterm | 0.510 | 0.650 | 0.494 | 3737 |
| Red group share | 0.356 | 0.609 | 0.499 | 2441 |
| Blue/Yellow group share | 0.644 | 0.391 | 0.499 | 2441 |
| Number of academies | 22 | 20 | – | 42 |

Notes: Means and SDs at student level. Follow-up indicators equal 1 if observed in that wave.

Table A4: Baseline Balance

| | Experiment 1 (Kenya) | | | Experiment 2 (Liberia) | | |
|---|---|---|---|---|---|---|
| | Control#mean (1) | Coef. (2) | N (3) | Control#mean (4) | Coef. (5) | N (6) |
| Class size | 19.094 | -3.949*** | 213 | 22.357 | 2.652 | 99 |
| | | (1.261) | | | (2.497) | |
| Lower grade class size | 19.697 | -3.563** | 212 | 23.286 | 1.377 | 99 |
| | | (1.387) | | | (2.630) | |
| Upper grade class size | 18.528 | -4.370*** | 213 | 21.224 | 3.158 | 99 |
| | | (1.292) | | | (2.830) | |
| Baseline test score | 0.037 | 0.040 | 213 | -0.061 | -0.150 | 99 |
| | | (0.074) | | | (0.118) | |
| Lower grade baseline score | 0.010 | 0.096 | 212 | 0.030 | -0.127 | 99 |
| | | (0.098) | | | (0.155) | |
| Upper grade baseline score | 0.066 | -0.018 | 213 | -0.120 | -0.198 | 99 |
| | | (0.075) | | | (0.128) | |

Notes: Kenya specifications are at the school level and control for constituency fixed effects, with standard errors clustered at the school level. Liberia specifications are at the school grade group level and control for randomization strata fixed effects, with standard errors clustered at the school grade group level. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table A5: Nigeria Baseline Balance

| Variable | Treat mean | Control mean | Diff | SE | p value |
|---|---|---|---|---|---|
| bl_score | 80.171 | 82.608 | -3.009 | ( 2.570) | 0.248 |
| has_bl | 0.606 | 0.651 | -0.023 | ( 0.033) | 0.477 |
| has_t2_mte | 0.622 | 0.634 | -0.027 | ( 0.041) | 0.521 |
| has_t2_ete | 0.691 | 0.677 | -0.001 | ( 0.034) | 0.971 |
| has_t3_mte | 0.470 | 0.516 | -0.103 | ( 0.073) | 0.163 |
| has_t3_ete | 0.650 | 0.510 | 0.135 | ( 0.084) | 0.117 |

Notes: Follow-up indicators are 1 if student observed in that wave. Diff column uses OLS with constituency and grade FE, clustered by academy.

Table A6: Test Score Follow-up

| | Experiment 1 (Kenya) | | | Experiment 2 (Liberia) | | |
|---|---|---|---|---|---|---|
| | Control#mean | Coef. | N | Control#mean | Coef. | N |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Full sample | 0.674 | 0.047 | 6,206 | 0.585 | 0.007 | 4,784 |
| | | (0.036) | | | (0.060) | |
| Lower grade | 0.690 | 0.050 | 3,193 | 0.560 | 0.045 | 2,434 |
| | | (0.043) | | | (0.073) | |
| Upper grade | 0.657 | 0.035 | 3,013 | 0.613 | -0.029 | 2,350 |
| | | (0.056) | | | (0.079) | |

Notes: All specifications use controls for baseline score, grade dummies (where applicable) and randomization strata fixed effects. Standard errors are clustered at the school-grade-group level. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table A7: Nigeria Attrition by Term and Window

| Wave/Outcome | Treat retain | Control retain | Diff | SE | p value |
|---|---|---|---|---|---|
| t2_mte_math | 0.608 | 0.628 | -0.027 | ( 0.042) | 0.517 |
| t2_mte_num | 0.610 | 0.589 | 0.020 | ( 0.061) | 0.747 |
| t2_ete_math | 0.687 | 0.676 | -0.004 | ( 0.034) | 0.902 |
| t2_ete_num | 0.691 | 0.676 | -0.001 | ( 0.034) | 0.985 |
| t3_mte_math | 0.460 | 0.514 | -0.116 | ( 0.075) | 0.128 |
| t3_mte_num | 0.464 | 0.504 | -0.100 | ( 0.074) | 0.183 |
| t3_ete_math | 0.650 | 0.510 | 0.135 | ( 0.084) | 0.117 |
| t3_ete_num | 0.636 | 0.493 | 0.143 | ( 0.089) | 0.118 |

Notes: Retention indicators defined unconditionally for all students in sample. OLS with constituency and grade FE, clustered by academy.

Table A8: Sample Flow — Liberia

| | Experiment (Liberia) | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| Sample flow step | Control | Treatment | Total |
| All observations | 3420 | 4165 | 7585 |
| Analytic sample | 2249 | 2535 | 4784 |
| With baseline score | 2249 | 2535 | 4784 |
| With endline score | 1482 | 1672 | 3154 |

Notes: Counts shown by treatment status at each sample construction step in the cleaned analysis dataset. The analytic sample keeps students with a nonmissing baseline score and at least two baseline scored students in the assignment unit used to form peer and classroom measures.

Table A9: Sample Flow — Kenya

| | Experiment (Kenya) | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| Sample flow step | Control | Treatment | Total |
| All observations | 8271 | 1447 | 9718 |
| Analytic sample | 6117 | 985 | 7102 |
| With baseline score | 6117 | 985 | 7102 |
| With endline score | 4195 | 759 | 4954 |

Notes: Counts shown by treatment status at each sample construction step in the cleaned analysis dataset. The analytic sample keeps students with a nonmissing baseline score and at least two baseline scored students in the assignment unit used to form peer and classroom measures.

Table A10: Nigeria Sample Flow

| Step | Control | Treatment | Total |
|---|---|---|---|
| All observations | 1995 | 1742 | 3737 |
| Non-missing treat and academy | 1995 | 1742 | 3737 |
| With baseline score | 1299 | 1055 | 2354 |
| Observed in T2 midterm | 1265 | 1084 | 2349 |
| Observed in T2 endterm | 1350 | 1203 | 2553 |
| Observed in T3 midterm | 1029 | 819 | 1848 |
| Observed in T3 endterm | 1018 | 1133 | 2151 |

Notes: Sample-flow counts in the Nigeria cleaned wide analysis file.

## A.2 Nigeria Outcome Details

Table A11: Effect on T2 Endterm Outcomes

| | (1) | (2) | (3) |
|---|---|---|---|
| | Math | Numeracy | Math + Numeracy Index |
| Treatment | 0.151 | 0.135 | 0.146* |
| | ( 0.115) | ( 0.083) | ( 0.082) |
| Control Mean | -0.000 | 0.000 | -0.000 |
| N | 2032 | 2037 | 2038 |

Notes: OLS with baseline score control, constituency FE, and grade FE. SEs clustered at academy level. ***, **, and * indicate 1%, 5%, and 10%.

Table A12: Effect on T3 Endterm Outcomes

| | (1) | (2) | (3) |
|---|---|---|---|
| | Math | Numeracy | Math + Numeracy Index |
| Treatment | 0.040 | 0.104 | 0.105 |
| | ( 0.150) | ( 0.144) | ( 0.132) |
| Control Mean | -0.000 | 0.000 | -0.015 |
| N | 1661 | 1616 | 1661 |

Notes: OLS with baseline score control, constituency FE, and grade FE. SEs clustered at academy level. ***, **, and * indicate 1%, 5%, and 10%.

Table A13: Effects Using Data Over Terms (Stacked)

| | (1) | (2) | (3) |
|---|---|---|---|
| | Math | Numeracy | Math + Numeracy Index |
| Treatment | 0.075 | 0.080 | 0.075 |
| | ( 0.090) | ( 0.081) | ( 0.076) |
| Control Mean | -0.000 | 0.000 | 0.002 |
| N (student-waves) | 7010 | 6905 | 7060 |

Notes: Pooled OLS on stacked student-wave observations with wave FE, baseline score control, constituency FE, and grade FE. SEs clustered at academy level. ***, **, and * indicate 1%, 5%, and 10%.

Table A14: Over-Term Effects by Group Color (Red/Blue/Yellow, Stacked)

| | Math | Numeracy | Math + Numeracy Index |
|---|---|---|---|
| Red group effect | 0.016 | 0.156 | 0.077 |
| | ( 0.104) | ( 0.104) | ( 0.091) |
| Blue group effect | 0.035 | 0.049 | 0.037 |
| | ( 0.121) | ( 0.116) | ( 0.109) |
| Yellow group effect | 0.306* | 0.127 | 0.225 |
| | ( 0.173) | ( 0.124) | ( 0.141) |
| N (student-waves) | 6741 | 6632 | 6783 |

Notes: Pooled OLS on stacked student-wave observations with wave FE. Rows report group-specific total treatment effects. Red is the treatment coefficient; Blue and Yellow are Treatment plus the corresponding interaction. SEs clustered at academy level. Controls as in baseline tables.

### A.3 Signal Quality and Assignment

Table A15: Baseline-Endline Predictive Power

| | Control-group diagnostics | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Study | Grade | N (Control) | Corr. | $R^2$ |
| Kenya | 1 | 2190 | 0.713 | 0.509 |
| | 2 | 2005 | 0.748 | 0.559 |
| Liberia | 1 | 411 | 0.247 | 0.061 |
| | 2 | 399 | 0.240 | 0.057 |
| | 3 | 324 | 0.184 | 0.034 |
| | 4 | 348 | 0.255 | 0.065 |

Notes: Correlation is the Pearson correlation between baseline and endline raw scores in the control group, by grade. $R^2$ is the squared correlation and measures the fraction of endline variance predicted by the baseline diagnostic.

Table A16: Alternative Signal-Quality Diagnostics

| | Experiments | |
|---|---|---|
| | (1) | (2) |
| | Kenya | Liberia |
| R-squared grade FE only | 0.011 | 0.059 |
| R-squared grade FE + baseline | 0.534 | 0.109 |
| Incremental R-squared | 0.523 | 0.050 |
| Within-grade rank persistence | 0.684 | 0.242 |

Notes: All computed in control group. Incremental R-squared = additional variance explained by adding baseline score to a grade FE only model. Rank persistence is the within grade Spearman correlation between baseline and endline score ranks, weighted by grade size.

Table A17: Nigeria Signal Quality (Incremental R-squared, Control Group)

| | T2 MTE | T2 ETE | T3 MTE | T3 ETE |
|---|---|---|---|---|
| R-squared grade FE only | 0.023 | 0.083 | 0.224 | 0.129 |
| R-squared grade FE + baseline | 0.238 | 0.299 | 0.332 | 0.247 |
| Incremental R-squared | 0.215 | 0.216 | 0.108 | 0.118 |
| Within-grade rank persistence | 0.471 | 0.489 | 0.377 | 0.388 |

Notes: Control group only. Incremental R-squared = variance explained by adding baseline score to a grade FE only model predicting the raw follow-up index (math+numeracy average). Rank persistence is the within grade Spearman correlation, weighted by grade size. Comparable to Kenya (0.523) and Liberia (0.050).

Table A18: Deterministic Assignment Cutoffs

| | Experiments | | | | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| | Kenya | | | Liberia | | |
| Grade | Cutoff | N treated | Rule error | Cutoff | N treated | Rule error |
| 1 | 40 | 527 | 0.000 | 23 | 637 | 0.000 |
| 2 | 35 | 458 | 0.000 | 23 | 662 | 0.000 |
| 3 | | | | 14 | 621 | 0.000 |
| 4 | | | | 14 | 615 | 0.000 |

Notes: The table reports score thresholds used to assign treated students to the upper reading track. Rule error is the share of treated students whose observed upper-track assignment differs from the published deterministic cutoff rule.

Table A19: Classroom Reallocation under Cross-Grade Grouping

| | Experiments | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| | Kenya | | Liberia | |
| | Control | Treat | Control | Treat |
| Class size | 20.88 | 17.99 | 32.19 | 42.16 |
| Within-class grade variance | 0.000 | 0.209 | 0.000 | 0.211 |
| Number of grades in class | 1.000 | 1.896 | 1.000 | 1.981 |
| Squared track target misfit | 1.352 | 0.419 | 1.519 | 0.639 |
| Peer mean EB ability | -0.006 | 0.254 | 0.009 | 0.032 |

Notes: Entries are arm-specific means in the structural analysis sample. Realized classroom margins use treated-track classes in treatment and grade classes in control. Squared track target misfit is the square of EB ability minus the track target.

Table A20: Nigeria: Classroom Reallocation under Ability Grouping

| | Control | Treatment |
|---|---|---|
| Class size | 30.2 | 27.2 |
| Within class grade variance | 0.000 | 0.445 |
| Number of grades in class | 1.000 | 2.609 |
| Within class baseline SD | 0.686 | 0.809 |
| Peer mean baseline ability | 0.098 | -0.109 |

Notes: Arm-specific means at the classroom level. Treatment classrooms are academy-by-group; control classrooms are academy-by-grade.

Table A21: Nigeria: Effect on Within-Class Dispersion

| | (1) Baseline | (2) T2 ETE | (3) T3 ETE |
|---|---|---|---|
| Treatment | 0.104* | -0.008 | 0.005 |
| | ( 0.056) | ( 0.039) | ( 0.029) |
| Control Mean | 0.547 | 0.452 | 0.411 |
| N | 2354 | 2038 | 1661 |

Notes: Outcome is absolute deviation of student score from classroom mean. Classroom = academy × group (treatment) or academy × grade (control). Baseline dispersion column omits baseline control. All columns include constituency and grade FE. SEs clustered at academy. ***, **, and * indicate 1%, 5%, and 10%.

Table A22: Treatment Effects by Track and Ability Tercile (Liberia)

| | | Liberia | | | |
|---|---|---|---|---|---|
| | | (1) | (2) | (3) | (4) |
| Track | Ability Bin | ITT | SE | p value | N |
| Lower | Bottom | -0.192 | ( 0.214) | 0.372 | 460 |
| | Middle | 0.150 | ( 0.283) | 0.599 | 459 |
| | Top | -0.055 | ( 0.238) | 0.819 | 393 |
| Upper | Bottom | -0.204 | ( 0.174) | 0.244 | 626 |
| | Middle | -0.472* | ( 0.242) | 0.055 | 606 |
| | Top | -0.354** | ( 0.150) | 0.021 | 610 |

Notes: Within-track ability terciles. OLS with strata FE and EB control. Standard errors are clustered at the school grade group level. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table A23: Nigeria: Treatment Effects by Two-Group Track Position and Within-Group Ability Tercile (T3 ETE)

| Group | Ability Bin | ITT | SE | p value | N |
|---|---|---|---|---|---|
| Red | Bottom | -0.229 | ( 0.186) | 0.228 | 249 |
| | Middle | 0.134 | ( 0.208) | 0.523 | 171 |
| | Top | 0.196 | ( 0.190) | 0.311 | 278 |
| Blue/Yellow | Bottom | 0.126 | ( 0.199) | 0.529 | 320 |
| | Middle | 0.168 | ( 0.148) | 0.263 | 354 |
| | Top | 0.355** | ( 0.150) | 0.025 | 239 |

Notes: Nigeria is grouped using the preferred two-group collapse: Red versus Blue/Yellow. Within-group ability terciles are based on baseline score. OLS with baseline score, constituency FE, grade FE, SEs clustered at academy. ***, **, and * indicate 1%, 5%, and 10%.

Table A24: ITT Heterogeneity by Distance from Assignment Cutoff (Liberia)

| | Liberia | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Quintile | Mean $\lvert s-c \rvert$ | ITT | SE | N |
| Q1 | 1.7 | -0.128 | ( 0.174) | 813 |
| Q2 | 5.0 | -0.064 | ( 0.157) | 636 |
| Q3 | 8.0 | -0.188 | ( 0.148) | 556 |
| Q4 | 11.3 | -0.383* | ( 0.198) | 604 |
| Q5 | 17.2 | -0.385* | ( 0.219) | 545 |
| T $\times$ $\lvert s-c \rvert$ | | -0.0253 | ( 0.0176) | 3154 |

Notes: Quintiles of absolute distance from the grade group assignment cutoff. OLS with strata FE and EB ability control. Standard errors are clustered at the school grade group level. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table A25: Effect on Within-Class Dispersion in Test Scores

| | Baseline Scores | | | Predicted Outcomes | | | Endline Scores | | |
|---|---|---|---|---|---|---|---|---|---|
| | Control mean (1) | Coef. (2) | N (3) | Control mean (4) | Coef. (5) | N (6) | Control mean (7) | Coef. (8) | N (9) |
| ***Panel A: Kenya*** | | | | | | | | | |
| Full Sample | 0.663 | -0.257*** | 7102 | 0.771 | -0.267*** | 7102 | 0.661 | -0.130*** | 4954 |
| | | ( 0.026) | | | ( 0.021) | | | ( 0.026) | |
| Lower Grade | 0.648 | -0.215*** | 3618 | 0.757 | -0.222*** | 3618 | 0.659 | -0.135*** | 2594 |
| | | ( 0.032) | | | ( 0.027) | | | ( 0.033) | |
| Upper Grade | 0.678 | -0.307*** | 3484 | 0.785 | -0.321*** | 3484 | 0.663 | -0.125*** | 2360 |
| | | ( 0.031) | | | ( 0.027) | | | ( 0.039) | |
| ***Panel B: Liberia*** | | | | | | | | | |
| Full Sample | 0.542 | -0.022 | 4784 | 0.669 | -0.042 | 4784 | 0.448 | 0.092** | 3154 |
| | | ( 0.029) | | | ( 0.034) | | | ( 0.042) | |
| Lower Grade | 0.529 | 0.029 | 2417 | 0.648 | 0.034 | 2417 | 0.421 | 0.059 | 1679 |
| | | ( 0.035) | | | ( 0.040) | | | ( 0.063) | |
| Upper Grade | 0.554 | -0.074 | 2367 | 0.689 | -0.115** | 2367 | 0.481 | 0.135** | 1475 |
| | | ( 0.046) | | | ( 0.054) | | | ( 0.053) | |

Notes: The outcome is the absolute deviation of each student's standardized score from the classroom mean. All specifications include randomization-strata fixed effects. The baseline score columns omit the baseline control because the outcome is constructed from baseline scores; the predicted-outcome and endline score columns control for standardized baseline score. Standard errors are clustered at the school level in Kenya and at the school grade group level in Liberia. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table A26: Treatment Effects on Lesson Completion Rate (Kenya)

| | Lesson completion rate | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Treat | 0.018 | 0.022 | 0.018 | 0.024 |
| | [0.014] | [0.015] | [0.016] | [0.017] |
| Teacher attendance | | 0.085*** | | 0.090*** |
| | | [0.029] | | [0.034] |
| Class size | | | -0.001 | -0.001 |
| | | | [0.001] | [0.001] |
| Control mean | 0.172 | 0.172 | 0.173 | 0.173 |
| Observations | 1128 | 1122 | 780 | 776 |

*Notes:* Each column reports the coefficient from a regression of lesson completion rate on treatment status. The unit of observation is a school–subject–grade cell. Lesson completion is the fraction of scripted literacy-revision lessons completed, recorded in NewGlobe's classroom management system. Columns 2–4 add controls for teacher attendance rate and reading class size. Control means are computed in each column's estimation sample. Standard errors are clustered at the school level in brackets. ***, **, and * indicate significance at 1%, 5%, and 10%.

## A.4 Peer/Rank and Classroom-Production Diagnostics

Table A27: Approximate Peer/Rank Diagnostic (Borusyak–Hull Control Function)

| | Experiments | |
|---|---|---|
| | (1) | (2) |
| | Kenya | Liberia |
| Peer mean EB ability | -0.133** | -0.048 |
| | ( 0.052) | ( 0.108) |
| N | 4954 | 3153 |

Notes: Dependent variable is standardized endline score. Peer mean EB ability is the mean of classmates' EB ability, excluding the focal student. This is the approximate control function implementation: controls include baseline-decile × treatment × grade FE and strata FE. Standard errors are clustered at the school level in Kenya and at the school grade group level in Liberia. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table A28: Nigeria: Approximate Peer/Rank Diagnostic

| | (1) T2 ETE Index | (2) T3 ETE Index |
|---|---|---|
| Leave-out peer mean | 0.087 | -0.109 |
| | ( 0.066) | ( 0.100) |
| N | 2036 | 1659 |

Notes: Leave-out peer mean is the standardized mean baseline score of all other students in the same classroom (academy × group for treated, academy × grade for control), excluding the focal student. Controls: baseline-decile × treatment × grade FE and constituency FE. SEs are clustered at academy. The estimate is an approximate peer/rank diagnostic, not a separately identified structural peer effect. ***, **, and * indicate 1%, 5%, and 10%.

Table A29: Harmonized Peer/Rank Diagnostic Across Experiments

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | Kenya | Liberia | Nigeria | Pooled |
| Peer baseline ability (leave-out mean) | -0.133** | -0.048 | -0.121 | -0.105** |
| | ( 0.052) | ( 0.108) | ( 0.098) | ( 0.046) |
| N | 4954 | 3153 | 1659 | 9766 |

Notes: This diagnostic applies a harmonized approximate Borusyak–Hull-style specification across the three experiments, with baseline-decile × treatment × grade FE and country-nested strata FE. The peer variable is harmonized leave-self-out baseline ability (`peer_eb` for Kenya/Liberia, `peer_lo_bl` for Nigeria). SEs are clustered at the harmonized assignment cluster. The pooled column is a diagnostic common-coefficient summary, not a structural peer-effect estimate.

Table A30: Kenya Peer/Rank Diagnostics: Approximate vs. Exact Borusyak–Hull Specifications

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| | Decile FE | Ventile FE | Exact CF | Exact recentered | Exact CF (simple) |
| Peer mean EB ability | -0.133** | -0.136** | 0.001 | -0.009 | -0.003 |
| | ( 0.052) | ( 0.053) | ( 0.085) | ( 0.083) | ( 0.075) |
| Exact $\mu_i^{BH}$ control | | | -0.177* | | -0.166* |
| | | | ( 0.096) | | ( 0.088) |
| Own-score controls | Cell FE | Cell FE | Cubic EB × T × grade | Cubic EB × T × grade | Linear EB × T × grade |
| Strata FE | Yes | Yes | Yes | Yes | Yes |
| School-clustered SE | Yes | Yes | Yes | Yes | Yes |
| Observations | 4954 | 4954 | 4952 | 4952 | 4952 |

*Notes:* Columns (1)–(2) report the approximate Kenya peer/rank diagnostic specifications using baseline score bins × treatment × grade fixed effects. Column (3) is the exact Borusyak–Hull control function specification: it regresses standardized endline score on the realized peer mean, excluding the focal student, and the exact expected peer regressor $\mu_i^{BH} = p_j P_i(1) + (1 - p_j) P_i(0)$, where $P_i(1)$ is the peer mean under the deterministic treated reading-group rule and $P_i(0)$ is the peer mean under grade-based control classrooms. Column (4) uses the exact recentered regressor $\tilde{P}_i = P_i^{obs} - \mu_i^{BH}$. Column (5) keeps the exact control function but replaces the flexible cubic own-score controls with a simpler linear EB × treatment × grade control set. All exact specifications include strata FE and clustered standard errors at the school level. The probability $p_j$ is the constituency-specific assignment probability from the full Kenya year-1 randomization roster (50 of 292 schools assigned to treatment). ***, **, and * indicate significance at 1%, 5%, and 10%.

Table A31: Peer/Rank Diagnostic and Classroom Shifts: Kenya

| | Kenya specifications | |
|---|---|---|
| | (1) | (2) |
| | Decile FE | Ventile FE |
| *Panel A: Peer/rank composite ($\hat{\zeta}$)* | | |
| Effect of 1 SD increase in peer mean | -0.133** | -0.136** |
| | ( 0.052) | ( 0.053) |
| Observations | 4954 | 4954 |
| *Panel B: Treatment-induced classroom shifts* | | |
| $\Delta$ mismatch, $\lvert\hat{\theta} - \hat{I}\rvert$ (C − T) | 0.421 | |
| $\Delta$ peer composition (T − C) | 0.277 | |
| $\Delta$ class size (T − C) | -3.1 | |
| $\Delta$ within class baseline SD (C − T) | 0.326 | |
| *Panel C: Diagnostic accounting* | | |
| Overall ITT | 0.031 ( 0.052) | |
| Peer/rank contribution ($\hat{\zeta} \times \Delta\bar{\theta}$) | -0.037 ( 0.019) | |
| Remainder | 0.067 ( 0.053) | |

*Notes:* Panel A reports the approximate Borusyak–Hull control function peer/rank coefficient from regressing standardized endline scores on peer mean ability, excluding the focal student, conditioning on baseline score bin × treatment × grade fixed effects and strata FE. Standard errors are clustered at the school level. Panel B reports mean treatment-control differences in classroom characteristics. Mismatch is measured as $\lvert\hat{\theta}_i^{EB} - \hat{I}_k\rvert$ where $\hat{I}_k$ is the population-level track mean. Panel C reports the implied peer/rank component under the approximate specification ($\hat{\zeta}\times$ observed peer shift) and the residual component of the ITT. The accounting is diagnostic; it is not a separately identified structural decomposition. Bootstrap SEs (1,000 replications, clustered at the school level) are in parentheses in Panel C.

Table A32: Peer/Rank Diagnostic and Classroom Shifts: Liberia

| | Liberia |
|---|---|
| | (1) |
| | Estimate |
| *Panel A: Approximate Borusyak–Hull peer/rank coefficient* | |
| $\hat{\zeta}_{Lib}$ | -0.048 ( 0.108) |
| Observations | 3153 |
| *Panel B: Treatment-induced classroom shifts* | |
| $\Delta$ absolute EB-to-target mismatch (C − T) | 0.348 |
| $\Delta$ peer composition (T − C) | 0.041 |
| $\Delta$ class size (T − C) | 11.4 |
| $\Delta$ within class baseline SD (C − T) | 0.058 |
| Overall ITT | -0.211 ( 0.135) |

*Notes:* Panel A reports the approximate Borusyak–Hull control function peer/rank coefficient using decile × treatment × grade FE. Standard errors are clustered at the school grade group level. Panel B reports treatment-control differences in classroom characteristics. The near-zero peer/rank coefficient and small classroom composition shift provide little evidence that the measured peer/rank diagnostic accounts for the negative Liberia ITT pattern.

Table A33: Nigeria: Peer/Rank Diagnostic and Classroom Shifts (T3 ETE)

| | |
|---|---|
| *Panel A: Peer/rank composite* | |
| Effect of 1 SD increase in peer mean | -0.109 |
| | ( 0.100) |
| *Panel B: Treatment-induced shifts* | |
| $\Delta$ within class dispersion (C − T) | -0.136 |
| $\Delta$ peer composition (T − C) | -0.242 |
| $\Delta$ class size (T − C, student-wtd.) | 2.9 |
| *Panel C: Diagnostic accounting* | |
| Overall ITT | 0.105 ( 0.132) |
| Peer/rank contribution ($\hat{\zeta} \times \Delta\bar{\theta}$) | 0.026 |
| Remainder | 0.079 |

Notes: Panel A reports the approximate Borusyak–Hull peer/rank coefficient from regressing the T3 ETE index on leave-out peer mean, conditioning on baseline-decile $\times$ treatment $\times$ grade FE and constituency FE, clustered at academy. Panel B reports student-weighted mean treatment-control differences in the T3 accounting sample. Classroom-level means are reported separately in Table A20. Panel C reports the implied peer/rank component under the approximate specification and the residual component of the ITT. The accounting is diagnostic; it is not a separately identified structural decomposition.

Table A34: Approximate Peer/Rank Diagnostic by Score Density

| | Score-density tercile | | |
|---|---|---|---|
| | (1) Low (tails) | (2) Medium | (3) High (middle) |
| *Panel A: Approximate peer/rank estimate by density tercile* | | | |
| Peer mean EB ability ($\hat{\zeta}$) | -0.179** | -0.099* | -0.123** |
| | ( 0.081) | ( 0.053) | ( 0.062) |
| N | 1708 | 1672 | 1574 |
| | Pooled interaction | | |
| *Panel B: Interaction specification* | | | |
| Peer mean EB ability ($\hat{\zeta}$) | -0.141 | | |
| Peer EB $\times$ density interaction | -0.024 | | |
| | ( 0.039) | | |

*Notes:* Panel A reports the approximate Borusyak–Hull control function peer/rank coefficient ($\hat{\zeta}$) separately for terciles of the baseline score density evaluated at each student's own score. Students in the tails of the distribution (low density) have few classmates at similar ability levels, so their within class rank is insensitive to changes in peer composition. Students in the middle (high density) have many nearby classmates, so rank shifts more with peer changes. In the stylized rank model, $\hat{\zeta} \approx \pi + \eta \cdot (-f(\theta_i))$, where $\pi$ is the mean peer effect, $\eta$ is the rank effect, and $f(\theta_i)$ is the score density. If rank effects matter in this reduced-form composite ($\eta \neq 0$), $\hat{\zeta}$ would tend to be more negative in the high-density tercile, where rank is more responsive to peer shifts. Panel B reports a pooled interaction of peer mean with standardized density; the interaction is a diagnostic for whether the peer/rank coefficient changes with local score density. All specifications include baseline-decile $\times$ treatment $\times$ grade FE and strata FE, so they should be read as approximate cell-control specifications rather than exact design-based recentering checks. Density is estimated using a Gaussian kernel with Silverman bandwidth, computed within grade, and standardized to mean zero and unit variance within grade. Standard errors are clustered at the school level. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table A35: Nigeria: Peer/Rank Diagnostic by Score Density (T3 ETE)

| | Score-density tercile | | |
|---|---|---|---|
| | (1) Low (tails) | (2) Medium | (3) High (middle) |
| *Panel A: Approximate peer/rank diagnostic by density tercile* | | | |
| Peer mean ($\hat{\zeta}$) | 0.884** | -0.173 | -0.048 |
| | ( 0.351) | ( 0.122) | ( 0.113) |
| N | 27 | 726 | 906 |
| | Pooled interaction | | |
| *Panel B: Interaction specification* | | | |
| Peer mean ($\hat{\zeta}$) | -0.205 | | |
| Peer × density ($\approx \hat{\eta}$) | 0.134 | | |
| | ( 0.066) | | |

Notes: Panel A reports the approximate peer/rank diagnostic by tercile of baseline score density. Panel B interacts peer mean with standardized density. All specifications include baseline-decile × treatment × grade FE and constituency FE. SEs are clustered at academy. These estimates are used as diagnostics for residual classroom-production channels. ***, **, and * indicate 1%, 5%, and 10%.

## A.5 Robustness

Table A36: ITT Estimates under Alternative Specifications

| | Kenya | | Liberia | |
|---|---|---|---|---|
| | Coef. | SE | Coef. | SE |
| Baseline (EB + strata FE) | 0.031 | ( 0.053) | -0.212 | ( 0.135) |
| Raw baseline score control | 0.031 | ( 0.053) | -0.212 | ( 0.135) |
| No baseline control | 0.110 | ( 0.071) | -0.240* | ( 0.139) |
| Wild cluster bootstrap $p$ value | 0.622 | | 0.126 | |
| Permutation $p$ value | 0.573 | | 0.128 | |
| N | 4954 | | 3154 | |

*Notes:* Each row reports the ITT estimate of treatment on standardized endline scores under a different specification. The baseline controls for empirical Bayes ability and strata fixed effects, with standard errors clustered at the school level in Kenya and at the school grade group level in Liberia. The raw baseline score row replaces the EB shrinkage estimate with the standardized raw baseline score. Wild cluster bootstrap $p$ values use 999 replications (Cameron et al. 2008). Permutation $p$ values are from a Fisher-exact randomization test with 999 permutations within strata at the country-specific assignment unit: school in Kenya and school grade group in Liberia. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table A37: Nigeria ITT Robustness Across Specifications

| | T2 MTE | T2 ETE | T3 MTE | T3 ETE |
|---|---|---|---|---|
| Baseline + FE + BL control | -0.042 | 0.146* | 0.071 | 0.105 |
| | ( 0.101) | ( 0.082) | ( 0.107) | ( 0.132) |
| No baseline control (FE only) | -0.114 | 0.090 | 0.051 | 0.017 |
| | ( 0.103) | ( 0.109) | ( 0.119) | ( 0.138) |
| Wild cluster bootstrap p value | 0.695 | 0.132 | 0.555 | 0.423 |
| Permutation p value | – | – | – | 0.493 |
| N | 1898 | 2038 | 1463 | 1661 |

Notes: Primary endpoint is T3 ETE (column 4). Wild bootstrap uses 999 replications. Permutation p values use 999 reassignments of academy treatment within constituency.

Table A38: Nigeria Lee Bounds by Assessment Wave

| | T2 MTE | T2 ETE | T3 MTE | T3 ETE |
|---|---|---|---|---|
| Control attrition rate | 0.366 | 0.323 | 0.484 | 0.490 |
| Treatment attrition rate | 0.378 | 0.309 | 0.530 | 0.350 |
| Difference (T-C) | 0.012 | -0.014 | 0.046 | -0.140 |
| OLS ITT | -0.042 | 0.146 | 0.071 | 0.105 |
| Lee lower bound | -0.074 | 0.146 | -0.100 | -0.126 |
| Lee upper bound | -0.042 | 0.180 | 0.071 | 0.298 |

Notes: Attrition is missing outcome in each wave-specific column. Lee bounds trim the lower-attrition arm to match attrition rates.

Table A39: Treatment Effects With and Without Class-Size Controls

| | Kenya | | Liberia | |
|---|---|---|---|---|
| | (1) No CS | (2) With CS | (3) No CS | (4) With CS |
| Lower track (Treatment) | 0.046 | 0.043 | -0.019 | -0.108 |
| | ( 0.137) | ( 0.138) | ( 0.181) | ( 0.180) |
| T × Upper | -0.023 | -0.023 | -0.328 | -0.237 |
| | ( 0.147) | ( 0.148) | ( 0.201) | ( 0.189) |
| Upper track total | 0.023 | 0.021 | -0.348** | -0.345** |
| | ( 0.044) | ( 0.044) | ( 0.154) | ( 0.151) |
| Class size | | -0.001 | | 0.003 |
| | | ( 0.002) | | ( 0.002) |
| N | 4954 | 4954 | 3154 | 3154 |

*Notes:* Odd columns reproduce the track-interaction specification without class size controls. Even columns add realized reading class size as a control. Upper track total is the sum of the Treatment and T × Upper coefficients. All columns control for EB ability and strata FE. Standard errors are clustered at the school level in Kenya and at the school grade group level in Liberia. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table A40: Nigeria: Treatment Effects With and Without Class-Size Controls (T3 ETE)

| | (1) No CS | (2) With CS |
|---|---|---|
| Lower track (Treatment) | 0.043 | -0.055 |
| | ( 0.155) | ( 0.153) |
| T × Upper | 0.194 | 0.405** |
| | ( 0.116) | ( 0.163) |
| Upper track total | 0.236* | 0.350** |
| | ( 0.132) | ( 0.151) |
| Class size | | 0.011 |
| | | ( 0.005) |
| N | 1611 | 1611 |

Notes: Upper = Blue/Yellow group under the preferred support-aware two-group collapse. Upper track total = Treatment + T × Upper. Controls: baseline score, constituency FE, grade FE. SEs clustered at academy. ***, **, and * indicate 1%, 5%, and 10%.

Table A41: Robustness to Score Ceiling Effects (Kenya)

| | (1) OLS | (2) Tobit | (3) Trimmed OLS |
|---|---|---|---|
| Treatment | 0.031 | 0.040 | 0.022 |
| | ( 0.053) | ( 0.058) | ( 0.057) |
| N | 4954 | 4954 | 4471 |
| Right-censored obs. | | 176 | |

*Notes:* Column (1) reproduces the baseline OLS ITT on standardized endline scores. Column (2) estimates a Tobit model on the raw endline composite (upper limit = 50 points), controlling for raw baseline score and strata FE; the coefficient and SE are divided by the control-group SD of raw endline scores for comparability with column (1). Column (3) re-estimates OLS after dropping students in the top baseline score decile. Columns (1) and (3) control for EB ability and strata FE; column (2) controls for raw baseline score and strata FE. Standard errors are clustered at the school level. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table A42: Nigeria Robustness to Ceiling Effects (T3 ETE Math)

| | (1) OLS | (2) Tobit | (3) Trimmed OLS |
|---|---|---|---|
| Treatment | 0.040 | 0.030 | 0.040 |
| | ( 0.150) | ( 0.155) | ( 0.150) |
| N | 1661 | 1661 | 1661 |
| Right-censored obs. | | 72 | |

Notes: Column (2) is Tobit on raw percent score with upper limit 100, normalized by control-group SD. Column (3) drops top baseline decile.

Table A43: Treatment Effect on Score Variance

| | Kenya | Liberia |
|---|---|---|
| *Panel A: Aggregate variance* | | |
| Control | 0.988 | 0.921 |
| Treatment | 0.915 | 0.997 |
| *Panel B: Treatment on total variance* | | |
| Treatment | 0.038 | 0.054 |
| | ( 0.073) | ( 0.210) |
| *Panel C: Treatment on within class variance* | | |
| Treatment | -0.228*** | 0.137* |
| | ( 0.052) | ( 0.069) |
| *Panel D: Treatment on between class variance* | | |
| Treatment | 0.239*** | -0.082 |
| | ( 0.050) | ( 0.199) |
| N | 4954 | 3154 |

*Notes:* Panel A reports raw variance of standardized endline scores by treatment arm. Panels B–D regress individual-level squared deviations on treatment, controlling for EB ability and strata FE. Total variance uses deviations from the sample grand mean; within class variance uses deviations from realized reading classroom means; between class variance uses the squared deviation of the classroom mean from the grand mean. Standard errors are clustered at the school level in Kenya and at the school grade group level in Liberia. ***, **, and * indicate significance at 1%, 5%, and 10%.

# B Structural Notation

Table A44: Nigeria Treatment Effect on Score Variance

| | T2 MTE | T2 ETE | T3 MTE | T3 ETE |
|---|---|---|---|---|
| *Panel A: Aggregate variance* | | | | |
| Control | 0.761 | 0.707 | 0.888 | 0.684 |
| Treatment | 0.722 | 0.664 | 0.952 | 0.555 |
| *Panel B: Treatment on total variance* | | | | |
| Treatment | -0.035 | -0.092 | -0.123 | -0.127 |
| | ( 0.075) | ( 0.103) | ( 0.142) | ( 0.086) |
| *Panel C: Treatment on within class variance* | | | | |
| Treatment | -0.041 | -0.008 | 0.052 | -0.003 |
| | ( 0.045) | ( 0.063) | ( 0.065) | ( 0.043) |
| *Panel D: Treatment on between class variance* | | | | |
| Treatment | 0.024 | -0.109 | -0.166 | -0.146* |
| | ( 0.087) | ( 0.070) | ( 0.105) | ( 0.081) |
| N | 1898 | 2038 | 1463 | 1661 |

Notes: Treatment classes are academy by `group_id`; control classes are academy by grade. Controls: baseline score, constituency FE, grade FE.

| Symbol | Definition |
|---|---|
| $\theta_i$ | Latent ability of student $i$ |
| $s_i$ | Observed baseline diagnostic score |
| $\mu_{g,m}$ | Mean latent ability in grade $g$ and market $m$ |
| $\rho_{m,g}$ | Signal quality in grade $g$ and market $m$ |
| $a_i$ | Posterior expectation of ability given $(s_i, g(i))$ |
| $K_m$ | Number of tracks in market $m$ |
| $I_{k,m}$ | Intended instructional target of track $k$ in market $m$ |
| $k_i^*$ | Intended track assignment under the cutoff rule |
| $k_i$ | Realized track assignment after implementation frictions |
| $\omega_m$ | Assignment execution in market $m$ (summary of regime mixture and misclassification) |
| $R_j$ | School-level implementation regime |
| $p_{m,r}$ | Probability of implementation regime $r$ in market $m$ |

| | |
|---|---|
| $M_{m,r}(k,\ell)$ | Probability that intended track $k$ is realized as track $\ell$ under regime $r$ |
| $\tau_m$ | Assignment-channel delivery fidelity in market $m$ |
| $D_i(a)$ | Delivered instructional content under regime $a$ |
| $J_{c,m}$ | Fallback classroom-responsive content in classroom $c$ |
| $P_i(a)$ | Leave-self-out mean peer ability under regime $a$ |
| $r_i(a)$ | Within-class percentile rank under regime $a$ |
| $S_i(a)$ | Composite social object, $P_i(a) + \omega_r r_i(a)$ |
| $q_{c,m}$ | Productive-time shifter for classroom $c$ in market $m$ |
| $N_c$ | Class size |
| $V_c^g$ | Within-class grade dispersion |
| $\lambda$ | Assignment-payoff scale in the estimable production map |
| $\varphi$ | Strength of the composite social channel before scaling by $(1-\tau_m)$ |
| $\omega_r$ | Rank weight inside the composite social object |
| $\delta_m$ | Market-specific baseline productivity shifter |
| $\chi_N$ | Effect of class size on productive time |
| $\chi_V$ | Effect of grade dispersion on productive time |
| $\Theta$ | Vector of structural parameters |
| $ATE_m$ | Average treatment effect in market $m$ |

# C Structural Estimation Diagnostics

This appendix reports the diagnostics behind the structural estimates in Section 7. The first table gives a local finite-difference diagnostic for the stage-4 production parameters. The local weighted Jacobian has numerical rank seven for the seven common production parameters, but it is ill-conditioned. The estimates should therefore be read as a structured mapping from observed implementation regimes to counterfactual regimes, rather than as sharp estimates of each production coefficient.

Table 9 in the main text reports the validation restrictions used before interpreting the counterfactuals.

Table A46 reports the outcome-free calibration from assignment-channel process evidence into the delivery-fidelity primitive.

Table A46: Delivery-Fidelity Calibration

| Market | Process evidence | Value | Calibration rule | Prior | $\hat{\tau}$ | Role |
|---|---|---|---|---|---|---|
| Kenya | Lesson-completion treatment-control difference | 0.025 | $\text{clip}(0.30 + 8\Delta LC,\ 0.05,\ 0.95)$ | 0.500 | 0.500 | Estimation |
| Liberia | Lesson-completion treatment-control difference | 0.011 | $\text{clip}(0.30 + 8\Delta LC,\ 0.05,\ 0.95)$ | 0.385 | 0.385 | Estimation |
| Nigeria | DI numeracy completion in treatment and control | T 0.262; C 0.261 | $\text{clip}(0.10 + 0.80LC_T + 2(LC_T - LC_C),\ 0.05,\ 0.85)$ | 0.312 | 0.312 | Estimation |
| Nigeria | Wrong-track / partial-script proxy | 0.350 | Not used in calibration | – | – | Validation only |

*Notes:* The table documents the outcome-free mapping from assignment-channel process evidence into the delivery-fidelity primitive. $\Delta LC$ is the treatment-control lesson-completion difference. $LC_T$ and $LC_C$ are DI numeracy completion rates in Nigeria treatment and control schools. The wrong-track proxy is retained as validation evidence and is not used to set $\tau$.

Table A47 re-estimates the production block under lower and upper process-to-$\tau$ mappings, holding the same structural target set fixed.

Table A47: Sensitivity to Delivery-Fidelity Calibration

| Spec. | $\tau_K$ | $\tau_L$ | $\tau_N$ | Max ITT/SE | K high-$\tau$ | High-input |
|---|---|---|---|---|---|---|
| Lower process map | 0.40 | 0.31 | 0.22 | 0.297 | +0.179 | +0.206 |
| Preferred | 0.50 | 0.38 | 0.31 | 0.281 | +0.155 | +0.187 |
| Upper process map | 0.60 | 0.46 | 0.40 | 0.276 | +0.126 | +0.165 |

*Notes:* Each row re-estimates the stage-4 production block after replacing the stage-3 delivery-fidelity primitives with a transparent lower, preferred, or upper process-to-$\tau$ mapping. The lower and upper mappings preserve the same process moments and market ordering logic but vary the intercept and slope used to translate completion evidence for the assignment channel into $\tau$. The high-input counterfactual still sets $\rho = \max_m \hat{\rho}_m$, $\omega = 0.98$, and $\tau = 0.95$; this table asks whether the observed-cell calibration of $\tau$ materially changes the extrapolation.

Table A48: Local Identification of Stage-4 Production Parameters

| Parameter | Role | Sensitivity norm | Largest local leverage |
|---|---|---|---|
| $\lambda$ | Assignment-payoff scale | 0.256 | Kenya: ITT |
| $\varphi$ | Social-channel scale | 2.132 | Kenya: Peer/rank |
| $\omega_r$ | Rank weight | 0.717 | Kenya: ITT |
| $\chi_N$ | Class-size pressure | 0.292 | Nigeria: ITT |
| $\chi_V$ | Grade-dispersion pressure | 0.165 | Nigeria: Dispersion |
| $\alpha$ | Delivery activation | 0.588 | Kenya: ITT |
| $\kappa$ | Sorting compression | 28.288 | Kenya: Dispersion |

*Notes:* The sensitivity norm is the Euclidean norm of finite-difference derivatives of weighted fitted stage-4 moments with respect to the raw optimization parameter at the preferred estimate. "Largest local leverage" reports the target moment with the largest absolute weighted derivative. The table is a local diagnostic, not an additional identifying assumption.

The next two tables report conditional uncertainty for the common production parameters and sensitivity to the weakly identified rank component inside the peer/rank composite. The delivery-activation parameter remains above one in the re-estimation draws, and fixing the rank weight to zero leaves the central counterfactual nearly unchanged.

Table A49: Stage-4 Production Parameter Uncertainty

| Parameter | Role | Point | 5th–95th pct. |
|---|---|---|---|
| $\lambda$ | Assignment-payoff scale | 0.400 | [0.400, 0.405] |
| $\varphi$ | Social-channel scale | 0.192 | [0.155, 0.216] |
| $\omega_r$ | Rank weight in social composite | -0.281 | [-2.000, 2.000] |
| $\chi_N$ | Class-size pressure | 0.033 | [0.008, 0.182] |
| $\chi_V$ | Grade-dispersion pressure | 0.022 | [0.010, 0.054] |
| $\alpha$ | Delivery activation | 4.317 | [1.202, 10.025] |
| $\kappa$ | Sorting compression | 1.649 | [1.644, 1.649] |

*Notes:* Percentiles use the same 49 converged parametric re-estimation draws of the stage-4 target moments used for the conditional counterfactual intervals. Blocks 1–3 primitives are held fixed, so intervals describe uncertainty in the production block conditional on measured $(\hat{\rho}, \hat{\omega}, \hat{\tau})$.

Table A50: Social-Channel Parameter Sensitivity

| Spec. | $\omega_r$ | Peer RMSE | Max ITT/SE | High-input | Nigeria $\rho + \tau$ |
|---|---|---|---|---|---|
| Preferred | -0.281 | 0.518 | 0.281 | +0.187 | +0.203 |
| No rank weight | 0.000 | 0.520 | 0.275 | +0.181 | +0.194 |

*Notes:* The second row re-estimates the stage-4 production block after fixing the rank weight inside the social composite to zero. Peer/rank RMSE is the root mean squared weighted error across the three peer/rank diagnostic targets. Max ITT/SE is the largest absolute fitted-minus-target ITT residual divided by the corresponding study-specific standard error. The diagnostic asks whether the main assignment-delivery counterfactual relies on separately estimating a rank component.

Table A51 reports the objective decomposition and optimizer diagnostics for the preferred production block. The hard comparative-static penalties are zero at the optimum, so the counterfactual is not produced by sitting on an inequality constraint. All local starts converge, and the high-input ATE is stable across starts, indicating that the headline counterfactual is not a numerical artifact of one optimizer initialization. Table A52 reports the exact non-moment terms in the preferred criterion.

Table A51: Stage-4 Objective and Optimizer Diagnostics

| Diagnostic | Value | Share (%) | Interpretation |
|---|---|---|---|
| *A. Objective decomposition* | | | |
| Moment fit | 0.956 | 57.8 | Weighted distance between targeted moments and model. |
| ITT moments | 0.150 | 9.1 | Contribution from the three country ATE targets. |
| Peer/rank moments | 0.804 | 48.6 | Contribution from the composite social-channel diagnostics. |
| Dispersion moments | 0.000 | 0.0 | Contribution from sorting-compression targets. |
| Assignment-payoff slope | 0.002 | 0.1 | Contribution from Kenya's revealed assignment-payoff target. |
| Market residual prior | 0.540 | 32.6 | Penalty for using country residuals to interpolate ITTs. |
| Auxiliary scale restrictions | 0.160 | 9.6 | Raw-parameter norm plus weak scale priors. |
| Hard comparative-static penalties | 0.000 | 0.0 | Zero means no acceptance inequality is binding. |
| *B. Numerical optimizer checks* | | | |
| Optimizer starts | 25 | | Multi-start L-BFGS-B runs. |
| Converged starts | 25 | | Starts reporting optimizer convergence. |
| Near-optimal starts | 18 | | Starts within $10^{-3}$ of the best objective. |
| Objective range | 1.655–1.677 | | Range across converged starts. |
| High-input ATE range | +0.185–+0.189 | | Range across converged starts. |

*Notes:* Panel A decomposes the preferred stage-4 criterion at the optimum. Objective shares are relative to the total objective. The hard comparative-static penalties are the inequality penalties used to rule out counterfactual mappings that violate the model's monotonicity and dominance restrictions. Panel B reports multi-start optimizer diagnostics; the high-input ATE is the fully idealized $\rho$-$\omega$-$\tau$ counterfactual.

Table A52: Stage-4 Normalizations

| Term | Objective term | Preferred setting | Obj. | Role |
|---|---|---|---|---|
| Market residual shrinkage | $\sum_m (\delta_m/0.25)^2$ | prior SD 0.25 | 0.540 | Prevents country intercepts from mechanically interpolating ITTs. |
| Raw-parameter norm | $0.05\,\mathrm{mean}(\tilde{\Theta}^2)$ | weight 0.05 | 0.153 | Numerical scale restriction on transformed parameters. |
| Assignment-payoff scale | $10[(\lambda - 0.4)/0.2]^2$ | center 0.40; scale 0.20 | 0.000 | Sets the preferred scale for high-fidelity assignment payoffs. |
| Social-channel scale | $2[(\varphi - 0.2)/0.15]^2$ | center 0.20; scale 0.15 | 0.006 | Keeps peer/rank scale in a plausible range. |
| Comparative-static penalties | monotonicity and dominance inequalities | hard penalties | 0.000 | Rules out counterfactual mappings that violate the theory. |

*Notes:* The table reports the non-moment terms in the preferred stage-4 minimum-distance objective. "Obj." is the contribution at the preferred estimate. The assignment-payoff and social-channel rows are scale normalizations; Appendix Table A54 reports sensitivity to relaxing them.

Tables A53–A60 report the main robustness battery. The residual, scale, delivery-calibration, target-block, and market-level checks ask whether the high-input counterfactual is a nuisance-intercept, calibration, single-moment, or single-market artifact. It is not: the preferred high-input ATE remains positive across these exercises, with the benchmark magnitude depending on auxiliary scale restrictions but not on any one target block, market, or observed-$\tau$ calibration.

The market-level exercise also clarifies scope. The Nigeria-environment $\rho + \tau$ decomposition is sensitive to omitting Nigeria's own stage-4 targets, so the paper treats that decomposition as mechanism evidence rather than as the headline counterfactual. The primitive-sensitivity and primitive-uncertainty tables ask whether the conclusion is driven by the benchmark values for $\rho$, $\omega$, and $\tau$. The combined-uncertainty table then pairs primitive draws with the stage-4 production-parameter draws; the fully high-input interval remains positive at the 5th percentile, while Nigeria-environment complementarity intervals are wider. The counterfactual is most sensitive to the high-delivery benchmark, but conservative joint primitive choices still imply a positive effect. The delivery-activation table shows that the nonlinear activation term is anchored by Kenya's revealed-payoff diagnostic and by the full stage-4 criterion: linear activation predicts a positive effect, but it more than doubles the criterion because it fails that revealed-payoff diagnostic, while the specifications that fit it imply gains close to the preferred estimate.

Table A53: Sensitivity to Market-Residual Discipline

| Specification | Residual prior SD | Max $\lvert\delta_m\rvert$ | Max ITT error / SE | High-input ATE |
|---|---|---|---|---|
| Strict | 0.15 | 0.092 | 0.553 | +0.183 |
| Preferred | 0.25 | 0.134 | 0.281 | +0.187 |
| Loose | 0.50 | 0.185 | 0.097 | +0.190 |
| Unrestricted | none | 0.500 | 0.003 | +0.194 |

*Notes:* Each row re-estimates the stage-4 production block holding $(\hat{\rho}, \hat{\omega}, \hat{\tau})$ fixed. "Max ITT error / SE" is the largest absolute fitted-minus-target ITT residual divided by the corresponding study-specific standard error. The high-input ATE sets $\rho$ to the highest observed signal quality, $\omega = 0.98$, and $\tau = 0.95$. The unrestricted specification removes residual shrinkage and hits the $\delta_m$ bound, so it is reported as a fit diagnostic rather than the preferred model.

Table A54: Sensitivity to Auxiliary Scale Regularization

| Specification | Raw norm | $\lambda$ prior | $\varphi$ prior | Max ITT/SE | High-input |
|---|---|---|---|---|---|
| Preferred | 0.05 | 10.0 | 2.0 | 0.281 | +0.187 |
| No raw-norm penalty | 0.00 | 10.0 | 2.0 | 0.292 | +0.191 |
| Weak scale priors | 0.05 | 2.5 | 0.5 | 0.273 | +0.180 |
| No lambda/phi priors | 0.05 | 0.0 | 0.0 | 0.283 | +0.242 |
| No auxiliary scale restriction | 0.00 | 0.0 | 0.0 | 0.289 | +0.068 |

*Notes:* Each row re-estimates the stage-4 production block holding the residual prior at its preferred value and varying only auxiliary scale regularization. "Raw norm" is the coefficient on the raw-parameter norm penalty; "$\lambda$ prior" and "$\varphi$ prior" are the weights on scale priors centered at 0.4 and 0.2, respectively. The high-input ATE sets $\rho$ to the highest observed signal quality, $\omega = 0.98$, and $\tau = 0.95$.

Table A55: Influence of Stage-4 Target Blocks

| Specification | Omitted target | Max ITT/SE | Kenya high-$\tau$ | Nigeria $\rho + \tau$ | High-input |
|---|---|---|---|---|---|
| Preferred | None | 0.281 | +0.155 | +0.203 | +0.187 |
| Drop Kenya ITT | Kenya ITT | 0.567 | +0.126 | +0.202 | +0.159 |
| Drop Liberia ITT | Liberia ITT | 1.415 | +0.146 | +0.195 | +0.177 |
| Drop Nigeria ITT | Nigeria ITT | 2.794 | +0.157 | −0.121 | +0.189 |
| Drop peer/rank | Peer/rank moments | 0.285 | +0.129 | +0.166 | +0.159 |
| Drop assign-payoff | Kenya assignment-payoff slope | 0.284 | +0.156 | +0.232 | +0.174 |
| Drop dispersion | Within-class dispersion moments | 0.281 | +0.155 | +0.203 | +0.187 |

*Notes:* Each non-preferred row re-estimates the stage-4 production block after omitting the indicated target block, holding the stage-1 to stage-3 primitive estimates fixed. "Max ITT/SE" evaluates fitted observed-country ATEs against the harmonized study-specific ITT estimates, even when an ITT target is omitted from estimation. The exercise is a local influence diagnostic: the preferred specification remains the full target set.

Table A56: Market-Level Influence of Stage-4 Targets

| Spec. | Omit | Holdout | Max fit | K high-$\tau$ | NG $\rho + \tau$ | High-input |
|---|---|---|---|---|---|---|
| Preferred | None | | 0.281 | +0.155 | +0.203 | +0.187 |
| Drop Kenya | Kenya | 0.398 | 0.398 | +0.175 | +0.266 | +0.188 |
| Drop Liberia | Liberia | 1.572 | 1.572 | +0.129 | +0.176 | +0.160 |
| Drop Nigeria | Nigeria | 2.809 | 2.809 | +0.152 | −0.130 | +0.183 |

*Notes:* Each non-preferred row re-estimates the stage-4 production block after omitting all targeted stage-4 moments from the indicated market, holding the stage-1 to stage-3 primitives fixed. Held-out ITT/SE is the absolute difference between the model-predicted observed-cell ATE and the harmonized reduced-form ITT for the omitted market, divided by that ITT's standard error. The exercise is a market-level influence diagnostic, not an alternative preferred specification.

Table A57: Sensitivity to High-Input Primitive Benchmarks

| Scenario | $\rho$ | $\omega$ | $\tau$ | High-input ATE |
|---|---|---|---|---|
| Conservative joint | 0.509 | 0.95 | 0.90 | +0.142 |
| Lower $\rho$ only | 0.509 | 0.98 | 0.95 | +0.179 |
| Lower $\omega$ only | 0.533 | 0.95 | 0.95 | +0.182 |
| Lower $\tau$ only | 0.533 | 0.98 | 0.90 | +0.152 |
| Preferred | 0.533 | 0.98 | 0.95 | +0.187 |
| Upper Kenya $\rho$ | 0.559 | 0.98 | 0.95 | +0.195 |

*Notes:* Each row evaluates the high-input counterfactual in the Kenya production environment while holding the estimated production parameters fixed. The conservative joint row uses the lower Kenya grade-specific signal-quality estimate, $\omega = 0.95$, and $\tau = 0.90$. The preferred row uses the paper's main high-input benchmark, $\rho = \max_m \hat{\rho}_m$, $\omega = 0.98$, and $\tau = 0.95$.

Table A58: Sensitivity to Primitive-Estimation Uncertainty

| Scenario | Point | 5th pct. | Median | 95th pct. |
|---|---|---|---|---|
| Kenya high delivery | +0.155 | +0.150 | +0.155 | +0.159 |
| Nigeria high delivery only | +0.115 | +0.105 | +0.115 | +0.127 |
| Nigeria high signal + delivery | +0.203 | +0.185 | +0.203 | +0.221 |
| Nigeria all three | +0.217 | +0.213 | +0.218 | +0.222 |
| Fully high-input cell | +0.187 | +0.181 | +0.187 | +0.192 |

*Notes:* This table propagates uncertainty in the stage-1 to stage-3 primitives while holding the preferred stage-4 production mapping fixed. Signal quality draws use a Fisher transformation of the control-group baseline–endline correlation; delivery-fidelity draws use the process-moment variances; Nigeria assignment-execution draws use a 0.08 SD implementation-error scale, while deterministic Kenya and Liberia execution is held fixed. The table is a primitive-uncertainty sensitivity check, not a full-system bootstrap.

Table A59: Combined Primitive and Production Uncertainty

| Scenario | Point | 5th pct. | Median | 95th pct. |
|---|---|---|---|---|
| Kenya high delivery | +0.155 | +0.027 | +0.130 | +0.214 |
| Nigeria high delivery only | +0.115 | −0.117 | +0.106 | +0.312 |
| Nigeria high signal + delivery | +0.203 | −0.046 | +0.180 | +0.391 |
| Nigeria all three | +0.217 | −0.036 | +0.193 | +0.407 |
| Fully high-input cell | +0.187 | +0.057 | +0.165 | +0.226 |

*Notes:* This diagnostic combines uncertainty in the stage-1 to stage-3 primitives with the stage-4 production-parameter re-estimation draws. Signal-quality, delivery-fidelity, and Nigeria assignment-execution draws use the same sampling rules as Table A58; production-parameter draws are sampled from the converged stage-4 target-moment re-estimations used in Table A49. The exercise is an approximate combined-uncertainty check, not a nonparametric bootstrap of the three original experiments.

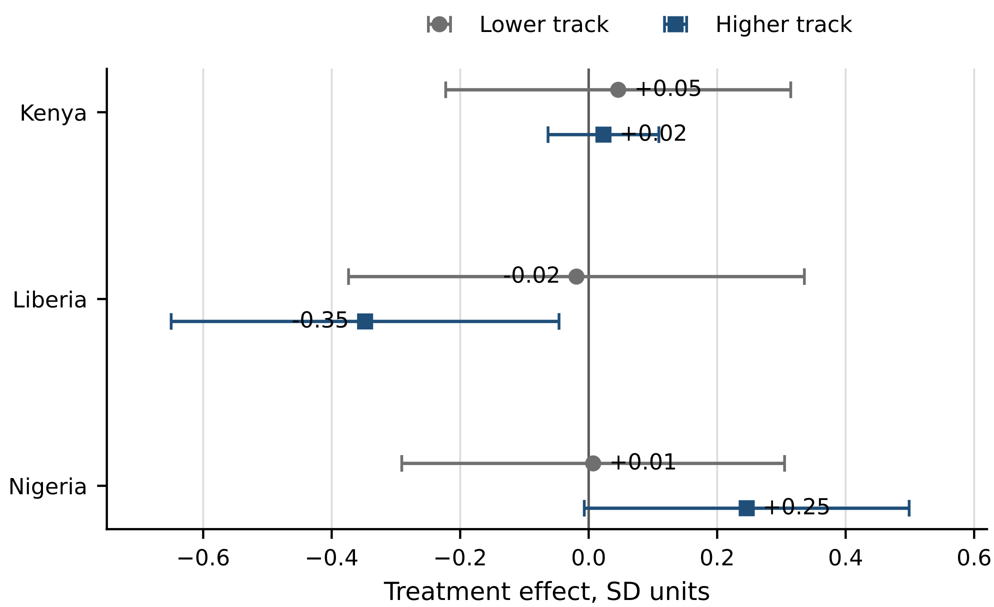
Table A60: Sensitivity to Delivery-Activation Functional Form

| Spec. | $\alpha$ | Obj. | Assign-payoff | Max ITT/SE | K high-$\tau$ | High-input |
|---|---|---|---|---|---|---|
| Linear | 1.00 | 3.648 | +0.101 | 0.286 | +0.120 | +0.127 |
| Quadratic | 2.00 | 2.071 | +0.053 | 0.285 | +0.152 | +0.168 |
| Cubic | 3.00 | 1.713 | +0.027 | 0.283 | +0.160 | +0.184 |
| Estimated | 4.32 | 1.655 | +0.011 | 0.281 | +0.155 | +0.187 |
| Steeper | 6.00 | 1.659 | +0.003 | 0.280 | +0.139 | +0.180 |

*Notes:* Each row re-estimates the stage-4 production block under the indicated delivery-activation exponent in $\tau^{\alpha}$, holding the stage-1 to stage-3 primitives fixed. The preferred row estimates $\alpha$; the other rows fix it. "Obj." is the full stage-4 minimum-distance criterion, including moment fit and regularization terms. "Assign-payoff fit" is the model-implied Kenya coefficient on treatment interacted with the predicted assignment-gain index, whose data target is +0.008. The table shows the fit–extrapolation tradeoff behind the delivery-activation restriction.

# D Supplementary Figure

Figure A1: Treatment Effects by Track Position Across Experiments




*Notes:* Points report country-specific treatment effects for lower-track and higher-track students, with 95% confidence intervals. The higher track is the upper reading group in Kenya and Liberia; in Nigeria it is the preferred two-group collapse, Blue/Yellow. The plotted estimates correspond to Table 17.